

**Angelo Cantoli - Simone Edmondo
Bedini**



POLITECNICO
MILANO 1863

Requirement Analysis and Specification Document

Deliverable:	RASD
Title:	Requirement Analysis and Verification Document
Authors:	Angelo Cantoli, Simone Edmondo Bedini
Version:	1.0
Date:	22/12/2025
Download page:	https://github.com/angelo-cantoli/CantoliBedini
Copyright:	Copyright © 2025 - Angelo Cantoli, Simone Edmondo Bedini - All rights reserved

Contents

Table of Contents	3
List of Figures	5
List of Tables	5
1 Introduction	6
1.1 Purpose	6
1.1.1 Goals	6
1.2 Scope	6
1.2.1 World Phenomena	6
1.2.2 Shared Phenomena	7
1.3 Definitions, Acronyms, Abbreviations	8
1.4 Revision History	8
1.5 Reference Documents	8
1.6 Document Structure	8
2 Overall Description	9
2.1 Product Perspective	9
2.1.1 Class Diagrams	9
2.1.2 State Diagrams	10
2.1.3 Scenarios	12
2.2 Product Functions	13
2.3 User Characteristics	15
2.4 Assumptions, Dependencies and Constraints	16
2.4.1 Domain Assumptions	16
2.4.2 Dependencies	16
3 Specific Requirements	17
3.1 External Interface Requirements	17
3.1.1 User interface	17
3.1.2 Hardware Interface	17
3.1.3 Software interface	17
3.1.4 Communication interface	17
3.2 Functional Requirements	17
3.2.1 Requirements	17
3.2.2 Mapping on goals	20
3.2.3 Use case diagrams	21
3.2.4 Use cases	22
3.2.5 Mapping on requirements	44
3.3 Performance Requirements	47
3.4 Design Constraints	47
3.4.1 Standards Compliance	47
3.4.2 Hardware Limitations	47
3.4.3 Any Other Constraint	47
3.5 Software System Attributes	48
3.5.1 Reliability	48
3.5.2 Availability	48
3.5.3 Security	48

3.5.4	Maintainability	48
3.5.5	Portability	48
4	Formal Analysis Using Alloy	49
4.1	Alloy Code	49
4.2	Simulations	54
5	Effort Spent	56
6	References	57
6.1	Reference Documents	57
6.2	Used Tools	57

List of Figures

1	Domain Model Class Diagram	9
2	Account Creation State Diagram	10
3	Manual Trip Insertion State Diagram	10
4	Automated Trip Recording State Diagram	11
5	Statistics Generation State Diagram	11
6	Trip Management State Diagram	12
7	High level Use Case Diagram	21
8	Use Case Diagram for UC1: Unregistered user signs up	23
9	Use Case Diagram for UC2: User logs in the account	25
10	Use Case Diagram for UC3: User manually adds bike trip	27
11	Use Case Diagram for UC4: User automatically records bike trip	29
12	Use Case Diagram for UC5: User views his bike trip history	30
13	Use Case Diagram for UC6: User views his bike trip statistics	31
14	Use Case Diagram for UC7: User publishes recorded trip	32
15	Use Case Diagram for UC8: User modifies one of his saved trips	34
16	Use Case Diagram for UC9: User views possible bike routes on map	35
17	Use Case Diagram for UC10: User views additional information about selected path on map	36
18	Use Case Diagram for UC11: User start navigation on selected bike path	37
19	Use Case Diagram for UC12: User modifies profile data	38
20	Use Case Diagram for UC13: User logs out the account	39
21	Use Case Diagram for UC14: User deletes the account	40
22	Use Case Diagram for UC15: User recovers password of the account	42
23	Use Case Diagram for UC16: User deletes a recorded trip	43
24	Simulation Steps 0-1	54
25	Simulation Steps 2-3	55
26	Simulation Step 4	55

List of Tables

1	Definitions, Acronyms and Abbreviations	8
4	Use Case UC1: Unregistered user signs up	22
6	Use Case UC2: User logs in the account	24
8	Use Case UC3: User manually adds bike trip	26
10	Use Case UC4: User automatically records bike trip	28
11	Use Case UC5: User views his bike trip history	30
12	Use Case UC6: User views his bike trip statistics	31
13	Use Case UC7: User publishes recorded trip	32
14	Use Case UC8: User modifies one of his saved trips	33
15	Use Case UC9: User views possible bike routes on map	35
16	Use Case UC10: User views additional information about selected path on map	36
17	Use Case UC11: User start navigation on selected bike path	36
18	Use Case UC12: User modifies profile data	37
19	Use Case UC13: User logs out the account	39
20	Use Case UC14: User deletes the account	40
21	Use Case UC15: User recovers password of the account	41
22	Use Case UC16: User deletes a recorded trip	43

1 Introduction

1.1 Purpose

BBP is a software designed to simplify the life of everyone who goes on bike trips and wants to enjoy them without worrying about finding the best routes.

It is intended as a community for cyclists, in which every person can share information about their personal routes, so that everyone may take advantage of the information and experience of other bikers. The application is intended for smartphone, where it can make use of GPS and sensor data to seamlessly reconstruct the path a cyclist has taken.

1.1.1 Goals

- G1: Cyclists maintain accurate and comprehensive records of their bike trips
 - G1.1: Bike trip data is automatically tracked while cycling
 - G1.2: Recording bike trips remains possible without automatic tracking
 - G1.3: The records of cyclists' bike trips are persistent and always accessible
 - G1.4: Cyclists' performance statistics are observable throughout their trip history
 - G1.5: Recorded bike trips include the condition of the roads on the bike path
 - G1.6: Hazards and obstacles on the bike path are identified and recorded
 - G1.7: Recorded bike trips include meteorological conditions when available
- G2: Cyclists make informed route selections
 - G2.1: All known paths that cover a route are observable
 - G2.2: The most suitable paths between two locations are distinguishable
 - G2.3: Path suitability is objectively evaluable based on road condition and route effectiveness
- G3: A community knowledge base of bike path information is maintained by cyclists
 - G3.1: Cyclists contribute bike path information by sharing their recorded trips
 - G3.2: Published bike path information is accessible to all users regardless of registration

1.2 Scope

1.2.1 World Phenomena

- WP1: User goes cycling along roads
- WP2: Roads contain potholes or other obstacles
- WP3: Road conditions vary over time
- WP4: Meteorological conditions change over time
- WP5: User needs to find routes between two locations
- WP6: User decides to record a bike trip
- WP7: Cyclists decide to share information with a community of other cyclists

1.2.2 Shared Phenomena

- SP1: User creates an account
- SP2: User logs into their account
- SP3: User logs out of their account
- SP4: User starts trip recording
- SP5: User stops trip recording
- SP6: System confirms trip has been saved
- SP7: System displays statistics regarding saved trips
- SP8: System retrieves meteorological information from an external service
- SP9: System displays trip information enriched with weather data
- SP10: User manually inserts street names and status of a bike path
- SP11: User manually inserts the presence of an obstacle on the path
- SP12: User enables the system to automatically acquire data from their mobile device
- SP13: System detects if the user is biking based on speed data
- SP14: Mobile device's GPS provides location coordinates for path reconstruction
- SP15: Mobile device's accelerometer provides motion data
- SP16: Mobile device's gyroscope provides orientation data
- SP17: User confirms automatically detected obstacles
- SP18: User corrects automatically detected obstacles
- SP19: User publishes bike trip information
- SP20: User specifies an origin and destination for path visualization
- SP21: System displays relevant path information
- SP22: System displays paths between two locations, ranked based on score
- SP23: User views their recorded trips
- SP24: User edits one of their recorded trips
- SP25: User deletes one of their recorded trips

1.3 Definitions, Acronyms, Abbreviations

Acronym	Definition
BBP	Best Bike Paths
GPS	Global Positioning System
UI	User Interface
SP	Shared Phenomenon
WP	World Phenomenon
UC	Use Case
DA	Domain Assumption
G	Goal

Table 1: Definitions, Acronyms and Abbreviations

1.4 Revision History

1.5 Reference Documents

- Assignment RDD AY 2025-2026.pdf
- Lecture slides and notes from the course Software Engineering 2 2025/2026
- Global Advisor-Cycling Across the World-2022 Report.pdf - <https://www.ipsos.com/sites/default/files/ct/news/documents/2022-05/Global>

1.6 Document Structure

The structure of the document is as follows:

- **Introduction:** The document begins by defining the purpose of BBP and outlining the core goals such as maintaining accurate trip records and enabling informed route selections.
This section also lists all phenomenas, differentiating between world phenomena and shared phenomena while providing essential definitions and acronyms.
- **Overall Description:** The second section provides a high-level perspective of the product through domain model class diagrams and state diagrams for account creation, trip insertion, and statistics generation.
It details primary functions, defining the characteristics of registered versus unregistered users and lists all assumption on the domain of the system and the relative environment.
- **Specific Requirements:** This technical section details the external interfaces for user, hardware, and software interactions. It enumerates the functional requirements covering every functionality of the app.
The core of this section consists of the use cases mapped to specific requirements and goals. It concludes with performance standards, design constraints and specification of system attributes.
- **Formal Analysis:** The fourth section utilizes the Alloy specification language to formally verify the system logic.
It includes the Alloy code that defines the lifecycle of a trip—from initial recording to verification and final publication, including constraints that specifies correct transitions between states.
- **Effort Spent:** The final section provides a transparent breakdown of the time invested by the authors, Bedini Simone and Cantoli Angelo.

2 Overall Description

2.1 Product Perspective

2.1.1 Class Diagrams

The following class diagram describes the basic components of the BBP system, their attributes and the relationship between them.

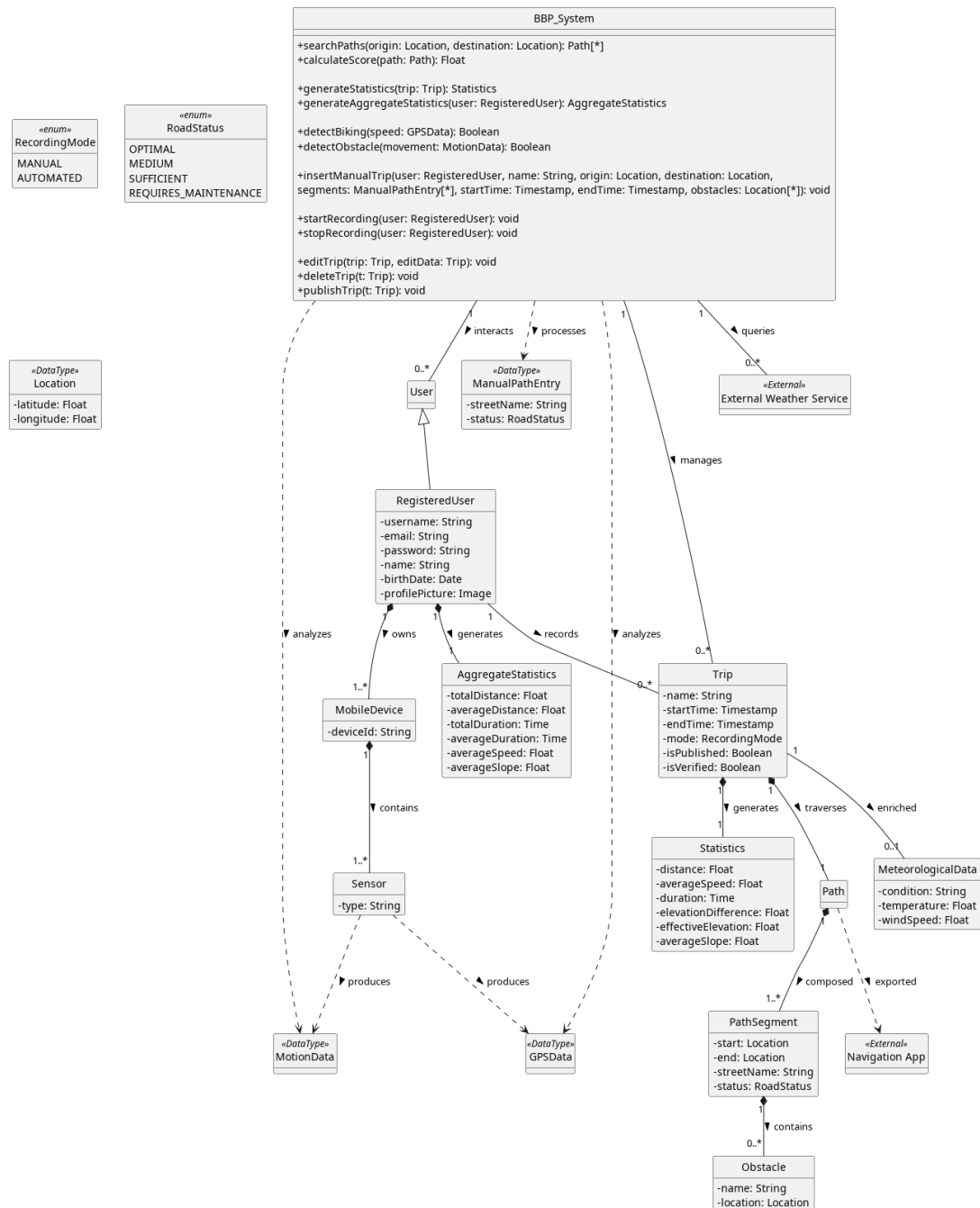


Figure 1: Domain Model Class Diagram

2.1.2 State Diagrams

Account Creation

The following state diagram describes the process of a new unregistered user creating a new account using the BBP app.

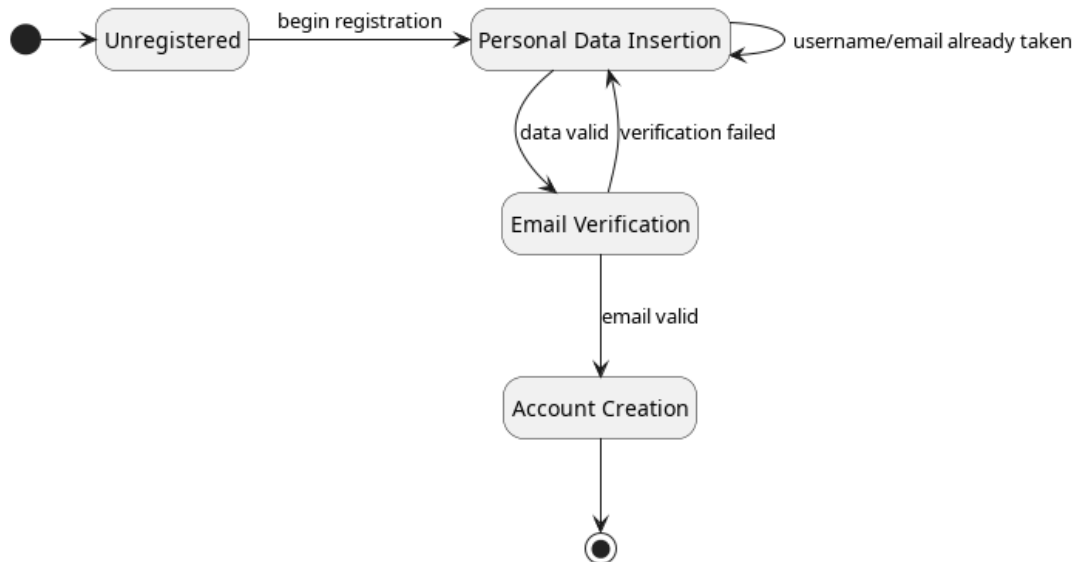


Figure 2: Account Creation State Diagram

Manual Trip Insertion

The following state diagram describes the process of a registered user manually inserting a new trip. The statistics generation process takes place after the insertion, and is described in the following state diagrams.

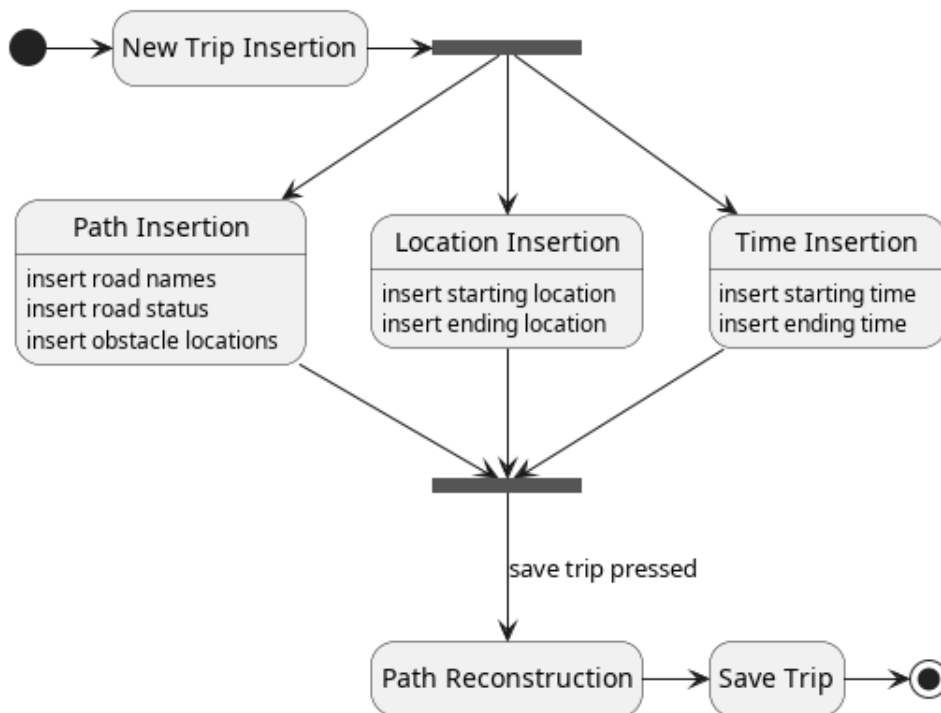


Figure 3: Manual Trip Insertion State Diagram

Automated Trip Recording

The following state diagram describes the process of a registered user recording a bike trip using BBP's automated recording feature. The statistics generation process takes place after the insertion, and is described in the following state diagrams.

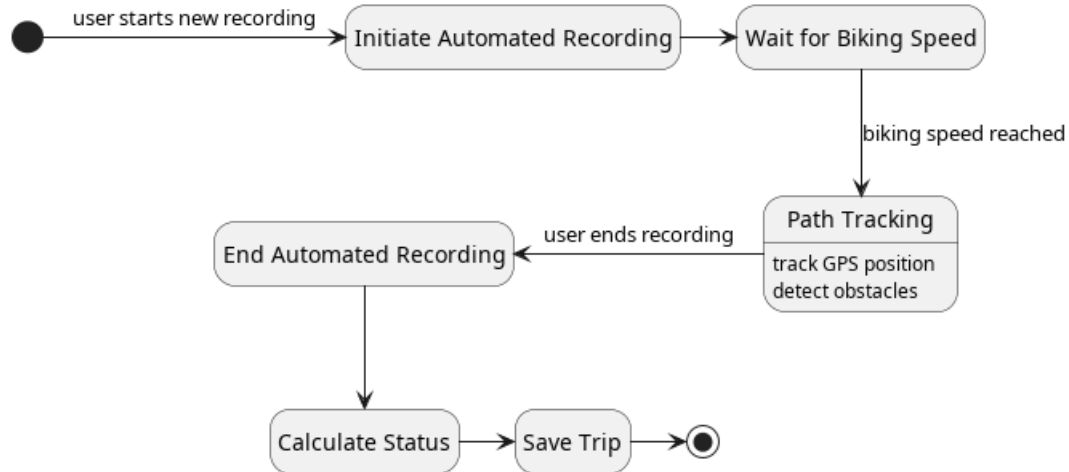


Figure 4: Automated Trip Recording State Diagram

Statistics Generation

The following state diagram describes the process of generating individual and aggregate statistics upon a trip being saved. It also details the process of enriching the trip data with meteorological information.

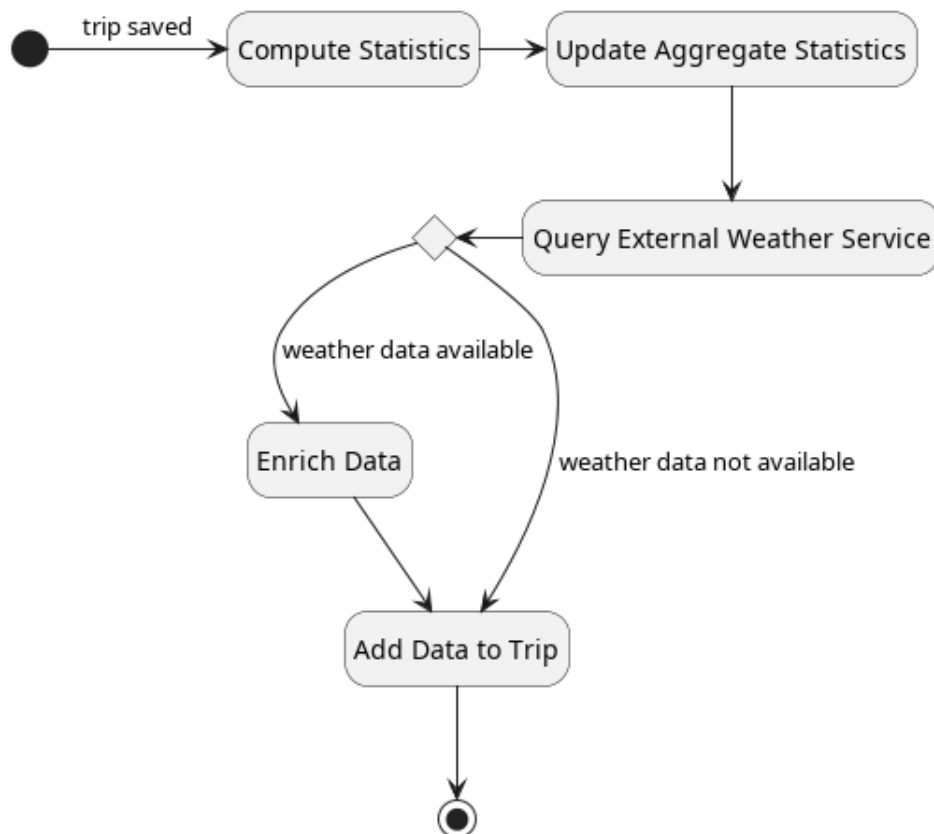


Figure 5: Statistics Generation State Diagram

Trip Management

The following state diagram describes the process of a registered user viewing and managing trips in their trip history.

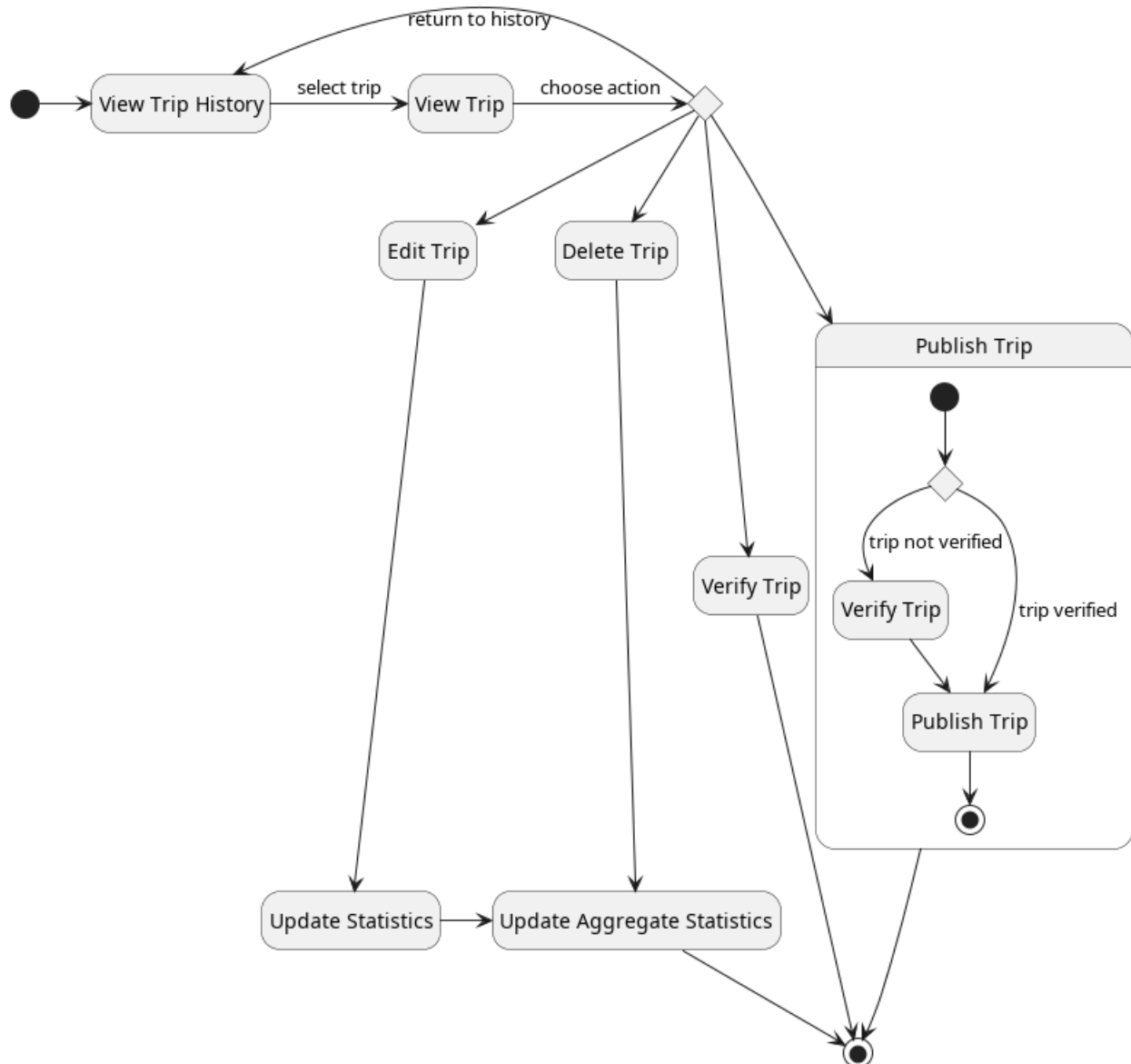


Figure 6: Trip Management State Diagram

2.1.3 Scenarios

User Creates an Account

J. Doe is a frequent cyclist, and desires an application that gives them the opportunity to log their cycling activity, and find information about the best bike routes in his city. They find that BBP perfectly suits their needs, and proceed to install the application. The application at first only shows the path viewing functionality: J. Doe wants to also record their trips, so they proceed to the 'sign up' section. The system then asks them to set a username, email and password, which J. Doe inserts, along with some optional personal information that they choose not to include. Inserting the username 'JD', the system responds that this username has already been taken, and prompts to insert a new username. Upon insertion of a different username, which was not taken by any other user, the system notifies that an email was sent

to the specified address with a code, which is required to complete the registration. Upon inserting the code, the system creates the new account, and J. Doe can begin using the complete application.

Registered User Automatically Records a Trip

J. Doe is about to go biking in their city. Before going out to get their bike, they decide that they want to record this trip using BBP, so they open the app, where they are logged in to their account, and initiate an automated recording of a new trip. The system starts acquiring GPS and sensor data from the smartphone, and waits until J. Doe starts biking: at that point, the speed detected by the system is high enough to start tracking the device in order to reconstruct the path. While biking, J. Doe encounters a pothole: their bike lurches to one side, and the system perceives from the mobile device's sensors that an abnormal movement has occurred, therefore it notes that an obstacle was encountered in that location. At the end of their trip, J. Doe gets off their bike and stops the recording, at which BBP notifies that the trip was successfully saved to their trip record.

Registered User Manually Records a Trip

J. Doe has finished cycling along a new route they found in their city, and they want to save the path in BBP while it is still fresh in their memory, since they didn't activate the automated recording before starting to bike. They go to the 'manual trip insertion' section of the application, which requests the insertion of a starting and ending location and time, and the names and status of all roads on the route, in the order in which they were traversed. Additionally the system gives the option to mark any obstacles encountered along the route: J. Doe remembers seeing some potholes on some of the roads, so they mark them. Once all the information has been inserted, they press 'save trip', at which point the system reconstructs the path from the provided data and saves the trip to J. Doe's record.

User Looks at Possible Bike Paths

J. Doe wants to bike to a new attraction that has recently opened in their city. However, he has never been to that neighbourhood, so they want to see if they can find any paths that can quickly and easily take them there. They open the BBP and go to the 'path viewing' section: after inserting their current location as an origin, and the attraction's location as a destination, the system searches for any published paths which are suitable for this route. It finds a couple of paths, and calculates a score for each one before displaying them on a map: the one with the highest score is highlighted. J. Doe selects the highlighted path and checks what additional information is provided: BBP shows that the path isn't the most direct way to reach the destination, but it is relatively flat and devoid of obstacles. J. Doe then selects the next highest scoring path: this path is more direct, but has more obstacles and roads in worse conditions, thus scoring lower. J. Doe considers that they want to reach the destination as quickly as possible, and thinks that their mountain bike is sturdy enough to not care too much about the worse road conditions, therefore they press the 'export path' button and choose to export the path to Google Maps, so they may begin navigation.

2.2 Product Functions

Sign up & Login

Users can access the app in two ways:

- Unregistered access: accessing the app without registering an account gives a user access to the path visualization interface. By inserting a start and end location, the application will show on a map all the paths between the two points, sorted based on their score (as described in detail in the 'Path Visualization' product function).
- Registered access: the app also gives users the option to register a personal account. It will ask the user to insert a username, e-mail and a password, along with some optional personal data (name,

date of birth, profile picture). If the username or email is already taken by another user, the system will notify that a different username must be chosen. A verification email will be sent to the provided address with an OTP, which will be requested by the system before the registration can be submitted. Once the registration is complete, the user can login to their account, which gives them the ability to record, and eventually publish, their bike trips, in addition to the features of the unregistered access. The user can log out from their account at any time.

Recording of Trips

The main functionality of BBP is to allow registered users to record and save their bike trips. There are two ways to record a trip in BBP:

- **Manual recording:** if the user wants to record their trip manually, they must insert all relevant information into the BBP application: the start and end locations of their trip, the names of the streets present in the path (in the order in which they were traversed), their status (selected from predefined status values: optimal, medium, sufficient, requires maintenance), and the start and end times of the trip. Users can also manually mark the locations of obstacles encountered along the path. The system will reconstruct the path based on the streets provided and their order.
- **Automatic recording:** if the user wants to record their trip automatically, they must begin the automated recording before the start of their trip, and allow BBP to acquire data from their mobile device. BBP will then start tracking the GPS position of the user's mobile device to determine what path they are taking; the tracking will begin once the speed of the mobile device is estimated to be elevated enough to assume that they have started biking (e.g. 15 km/h), and will not end until the users manually halts the recording. BBP will simultaneously access some of the mobile device's sensors, such as the gyroscope or the accelerometer, so that if a notable movement is registered (due to the presence of some obstacle on the road, such as a pothole) an obstacle marker is added to that point in the path. The system will also determine the status of the road based on the number of obstacles and sensor data patterns. Once the trip has ended the user must stop the recording, at which point BBP will save the trip.

The recorded trips are saved in a database accessible only to the user; trips can be published at any time so that other users may visualize them, however if the trip data was recorded automatically BBP will first ask the user to verify that the collected information is accurate, and if it is not to manually correct it by removing incorrect obstacle markers and adjusting the status of the roads, before publishing the trip.

Trip Statistics Generation

Once a user has recorded a trip, the system will automatically generate some statistics about the trip. The statistics include:

- total distance covered
- average speed
- time taken to complete the trip
- total elevation difference between start and end points
- effective elevation traversed
- average slope of the path

These statistics are calculated based on the recorded path data and the inserted start/end times. The system will also query an external service to obtain meteorological data relative to the path at that time: if it

is available, it will be added to the path statistics. The computed statistics are stored with the corresponding path in the database; the users have the option to access them when visualizing a path. Additionally, the system will compute aggregate statistics across all the trips the user has recorded (e.g. average time taken to complete trips, average elevation difference of trips, etc.)

Trip History Management

Users have the option to view all their past recorded trips. The trips are presented in a list format in chronological order, with the date of recording, the name and the start and end locations immediately visible. By selecting a specific trip, the user can view the computed statistics for that trip, and a map view of the path: there is an option to open the path in a third party navigation app (e.g. Google Maps), so that the user can navigate the same route again if they desire. The user has the option to edit or delete the trips at any time, and can also choose to publish the trip: if the data for the trip was automatically generated and the user has not yet reviewed it, the system will prompt the user to verify and correct the information before publishing. If a trip that has already been published is deleted or modified, the system will propagate the action to the published trip (e.g. if a user deletes a published trip from their personal trip history, the trip will no longer be available to other users); the user also has the option to make a published trip private again at any time, immediately removing it from public availability.

Path Visualization

Users, both registered and unregistered, can visualize published bike trips on a map. The application will prompt the user to insert a start and end location: once inserted, the system will search for all paths that connect those two points. The system then computes a score for each of the paths: the score is based on the status of the roads in the path, the time it would take for the user to reach the destination by following the path (calculated by considering the total length of the path and the elevation difference), the amount of obstacles along the path and the difficulty of the path (calculated by considering the average slope of the path). The system then shows the user all the identified paths on a map: the path with the highest score will be highlighted, while the others will be shown with lower opacity, and the scores will all be visible. Only the 5 paths with the highest score will be immediately visible, to avoid cluttering; the user will have the option to make all possible paths visible. The user can select a path by clicking on it: the application will then give the user the option to view the statistics of the path, select the path as the main highlighted path (if it isn't already), and to open the path in a third party navigation app (e.g. Google Maps). The user will also have the option to view all paths in a list, sorted by score.

2.3 User Characteristics

Unregistered User

An unregistered user can use BBP in a reading-only way, visualizing paths on the map and their score to receive suggestions on which bike trip to take. Furthermore at every moment he can decide to register to take advantage of all the features of BBP.

Registered User

A registered user is the main character of the app, and is fundamental for its correct functioning, because the database of routes of BPP is based on insertion of registered users. In fact, a registered user can visualize paths on the map like an unregistered user and they can also insert routes, manually or automatically, and make them public to let all users see them.

2.4 Assumptions, Dependencies and Constraints

2.4.1 Domain Assumptions

- DA1: Each registered user has a valid email address
- DA2: Users have a mobile device capable of running the BBP application
- DA3: Users' mobile devices have GPS capabilities, with typical smartphone precision (5-10 meters)
- DA4: GPS signals are generally available with sufficient precision while cycling
- DA5: Registered users who use the automated recording functionality have accelerometer and gyroscope sensors in their mobile devices
- DA6: Users' have internet access when using online features (e.g. path publication and visualization, weather data retrieval)
- DA7: Users bike on roads intended for cycling (where bike tracks exist, or cars are rare and speed limits are compatible with biking)
- DA8: Users who manually record trips bike on named roads which are marked on maps
- DA9: A sufficient number of registered users are willing to publish their bike routes
- DA10: Obstacles on bike paths cause a significant enough shift in movement to be detectable through the mobile device's sensors
- DA11: Users can assess road surface conditions and categorize them in the predefined status values
- DA12: Bike paths can be reconstructed reasonably accurately based on the names of the streets and the order in which they were traversed

2.4.2 Dependencies

Mapping and routing service

The BBP system depends on an external mapping and routing service (e.g GoogleMaps, OpenStreetMap, etc.) for tracking and visualizing the bike paths.

Meteorological service

The BBP system depends on an external, third-party meteorological service to acquire data relative to the weather conditions on the bike paths, which are then used to enrich the data about the path.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User interface

BBP's user interface is a mobile application, available for Android and iOS.

The UI should be intuitive and easy to understand, since the users will be of all age groups and with varying levels of familiarity with technology. The UI will contain sections to allow viewing bike paths from other users, viewing personal bike path archive, recording or inserting new bike paths, and publishing saved bike paths.

3.1.2 Hardware Interface

The system requires a smartphone with the BBP application installed and internet connection capabilities.

Furthermore, the smartphone should have a GPS localization system and some sensors, such as gyroscope and accelerometer: in absence of these the app will not be able to complete automated recordings of bike trips.

3.1.3 Software interface

The system requires some software interfaces for the correct functioning.

The most important is the DBMS that will be used for communicating with the database of the application, in fact many times data is required to be persistent so this interface is necessary to the correct functioning of the database.

Other than DBMS there are services focused on some specific feature of the application.

One is the third-party meteorological service and its API that make it possible for BBP to retrieve meteorological information and insert them in the application for enrichment of trips data.

Another important software is the third-party mapping service that is fundamental for one the most important features, to visualize paths on map, relative to published bike trips.

3.1.4 Communication interface

Some communication interfaces are involved in exchanging data between components in the network and clients.

The HTTP/HTTPS is fundamental to the front-end application to communicate with the back-end and to APIs of mapping and meteorological services. This protocol make possible and safe the hypertext transfer

JSON is another interface for communication, representing the format of the data payload transmitted over the network

Another communication interface is the SMTP protocol, for sending and receiving of email, useful for confirmation of email account sending OTP to user's email.

3.2 Functional Requirements

3.2.1 Requirements

Account registration and management

- R1: System accepts account registration data
- R2: System rejects account registrations with duplicate emails or usernames

- R3: System sends a verification email with an OTP to the email address provided during registration
- R4: System must verify OTP code before confirming registration
- R5: System allows users to login to their account by authenticating via email (or username) and password
- R6: System allows users to logout of their account
- R7: System allows users to modify their account information (username and personal data)
- R8: System allows users to modify/recover their password via email verification
- R9: System allows users to delete their account, with all the associated data

Automated trip recording

- R10: System allows users to start automated trip recording
- R11: System requests necessary sensor and GPS permissions when automated recording is started
- R12: System begins automated path and sensor tracking, after recording initiation, only when user has started biking
- R13: System algorithmically detects if a user has started biking from speed data
- R14: System collects GPS coordinates during automated trip recording
- R15: System reconstructs traveled path from collected GPS coordinates
- R16: System collects data from mobile device's accelerometer and gyroscope during automated trip recording
- R17: System algorithmically interprets collected sensor data to detect potential obstacles along the path
- R18: System allows users to manually stop automated trip recording

Manual trip recording

- R19: System allows users to manually record trips by inserting starting and ending locations, names of roads in traversal order, start and end times
- R20: System allows users to specify road status (optimal, medium, sufficient, requires maintenance) during manual recording
- R21: System allows users to specify obstacle locations during manual recording
- R22: System persistently stores recorded trips
- R23: System ensures only author of recorded trips can access unpublished trip records

Statistics calculation

- R24: System automatically calculates statistics for each recorded trip (distance, average speed, duration, elevation, average slope)
- R25: System calculates statistics immediately after trip is saved
- R26: System calculates aggregate statistics across all of user's recorded trips
- R27: System recomputes aggregate statistics upon modification of user's trip record (add/delete/edit trip)
- R28: System persistently stores computed statistics with corresponding trip data
- R29: System infers road status from detected obstacle density and movement pattern analysis when trip data is recorded automatically
- R30: System queries an external meteorological service for weather data to enrich trip data after completion
- R31: System stores a notice in trip data when weather data is unavailable

Recorded trip management

- R32: System displays a list of user's recorded trips showing name, date, start and end locations
- R33: System allows users to view detailed trip data for any recorded trip, including route map, statistics and weather data
- R34: System allows users to manually edit trip data for any of their recorded trips
- R35: System allows users to delete any of their recorded trips
- R36: System propagates modifications and deletions of trips to their published versions, if present
- R37: System allows users to make published trips private again

Trip publication

- R38: System allows registered users to publish their recorded trips, making them publicly available
- R39: System requires the user to correct and confirm data before publishing, if it was acquired automatically
- R40: System allows users to add/remove obstacle markers during data correction
- R41: System allows users to modify road status during data correction

Published path viewing

- R42: System allows all users to view published bike paths between specified start and end points
- R43: System identifies all published bike paths that connect specified start and end points
- R44: System notifies user when no published paths exist between the specified points
- R45: System calculates a score for each identified bike path based on road status, route effectiveness (distance and directness in reaching destination), amount of obstacles, path difficulty (slope)

- R46: System displays all identified paths on a map, with the highest scoring path highlighted and the others with reduced opacity
- R47: System displays score values for each path
- R48: System initially displays a limited amount of the identified paths, providing the option to view all paths
- R49: System allows users to select a displayed path
- R50: System allows users to view additional information about selected path (status, obstacles)
- R51: System allows users to export selected path to compatible third-party navigation applications

3.2.2 Mapping on goals

Table 2: Goal Mapping on Requirements

Goal	Domain Assumptions	Requirements
G1	DA1, DA2, DA3, DA4, DA5, DA6, DA7, DA8, DA10, DA11, DA12	R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35
G1.1	DA1, DA2, DA3, DA4, DA5, DA6, DA7, DA10	R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18
G1.2	DA1, DA2, DA6, DA7, DA8, DA11, DA12	R1, R2, R3, R4, R5, R6, R7, R8, R9, R19, R20, R21
G1.3	DA1, DA2, DA6	R1, R2, R3, R4, R5, R6, R7, R8, R9, R22, R23, R32, R33, R34, R35
G1.4	DA1, DA2, DA6	R1, R2, R3, R4, R5, R6, R7, R8, R9, R24, R25, R26, R27, R28
G1.5	DA1, DA2, DA5, DA6, DA7, DA10, DA11	R1, R2, R3, R4, R5, R6, R7, R8, R9, R16, R17, R21, R22, R29
G1.6	DA1, DA2, DA3, DA4, DA5, DA6, DA10	R1, R2, R3, R4, R5, R6, R7, R8, R9, R16, R17, R21, R22
G1.7	DA1, DA2, DA6	R1, R2, R3, R4, R5, R6, R7, R8, R9, R30, R31
G2	DA2, DA6	R42, R43, R44, R45, R46, R47, R48, R49, R50, R51
G2.1	DA2, DA6	R42, R43, R44, R46, R48, R49, R50, R51
G2.2	DA2, DA6	R43, R46, R47, R48, R50
G2.3	DA2, DA6	R43, R44, R45
G3	DA1, DA2, DA6, DA9	R1, R2, R3, R4, R5, R6, R7, R8, R9, R36, R37, R38, R39, R40, R41, R42
G3.1	DA1, DA2, DA6, DA9	R1, R2, R3, R4, R5, R6, R7, R8, R9, R36, R37, R38, R39, R40, R41
G3.2	DA1, DA2, DA6	R42

3.2.3 Use case diagrams

High level use case diagram

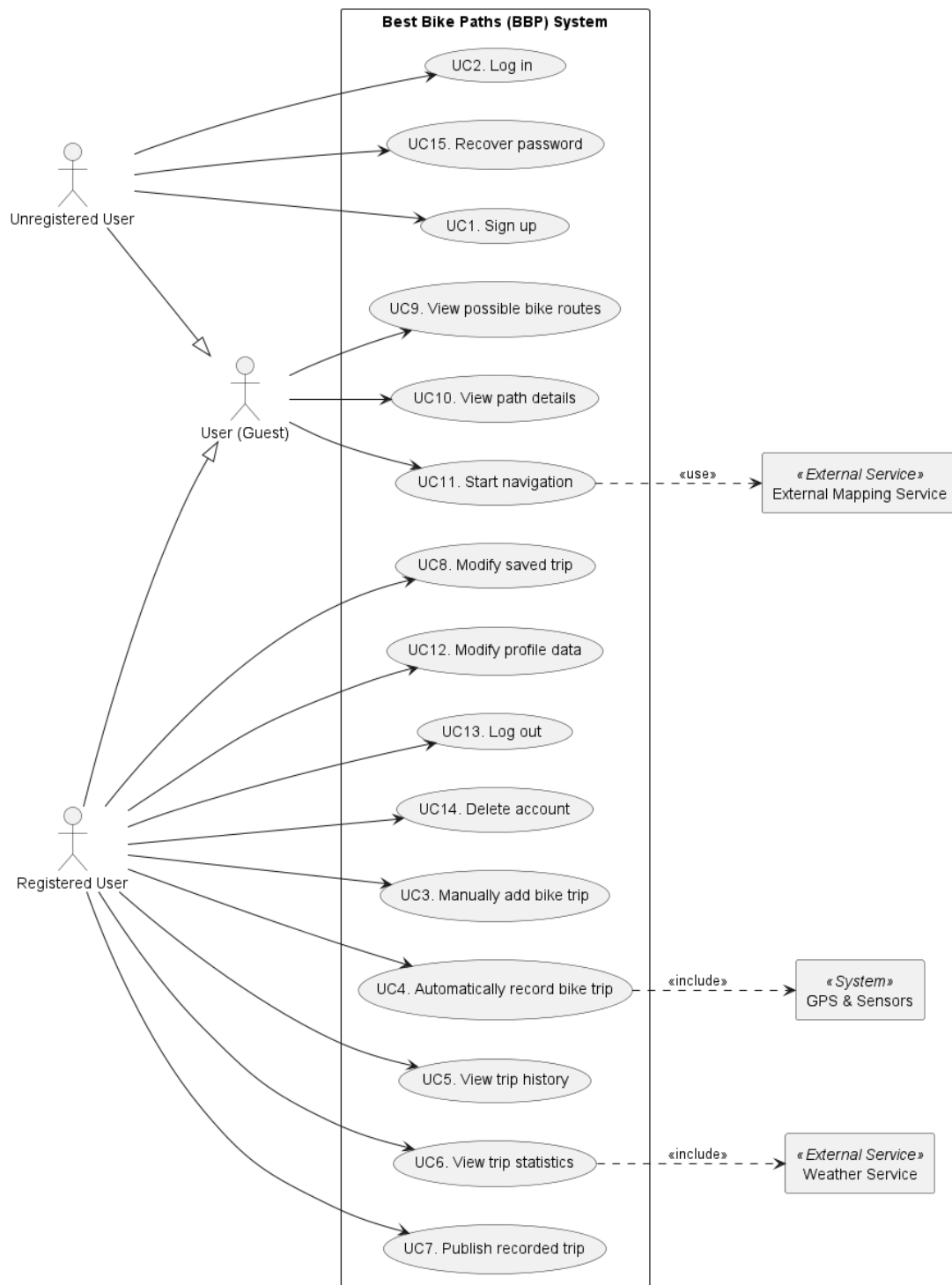


Figure 7: High level Use Case Diagram

3.2.4 Use cases

UC1. Unregistered user signs up

Actor	Unregistered user
Entry conditions	User is on login page of BBP
Event flow	1. User clicks to register on BBP 2. BBP asks user to insert his informations for the account 3. User inserts his informations 4. BBP validates the inserted personal informations 5. BBP sends a verification email to the provided address with an OTP 6. User inserts received OTP on BPP 7. BBP validates inserted OTP 8. BBP sends the registration outcome
Exit conditions	An account is created
Exceptions	Username already taken Password not valid Connection error while registering OTP not valid

Table 4: Use Case UC1: Unregistered user signs up

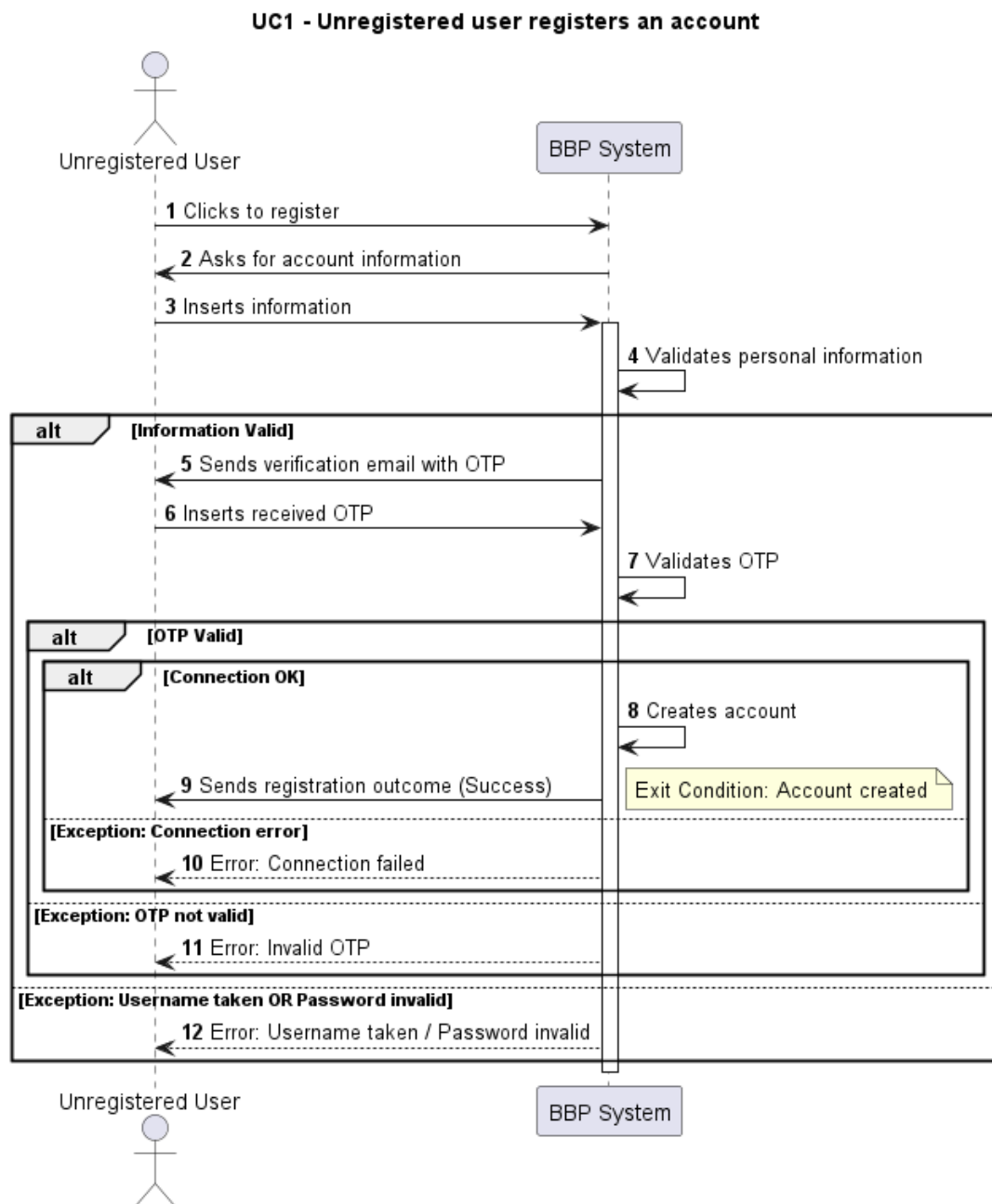


Figure 8: Use Case Diagram for UC1: Unregistered user signs up

UC2. User logs in the account

Actor	Registered user
Entry conditions	Unregistered is on login page of BBP
Event flow	1. User clicks to login on BBP 2. BBP asks user to insert username and password of the account he wants to log in 3. User inserts his username 4. BBP validates the inserted username and password 5. BBP sends login outcome
Exit conditions	User logs BBP system
Exceptions	Inserted username not existent Wrong inserted password for specified username Connection error while logging in

Table 6: Use Case UC2: User logs in the account

UC2 - User logs in the account

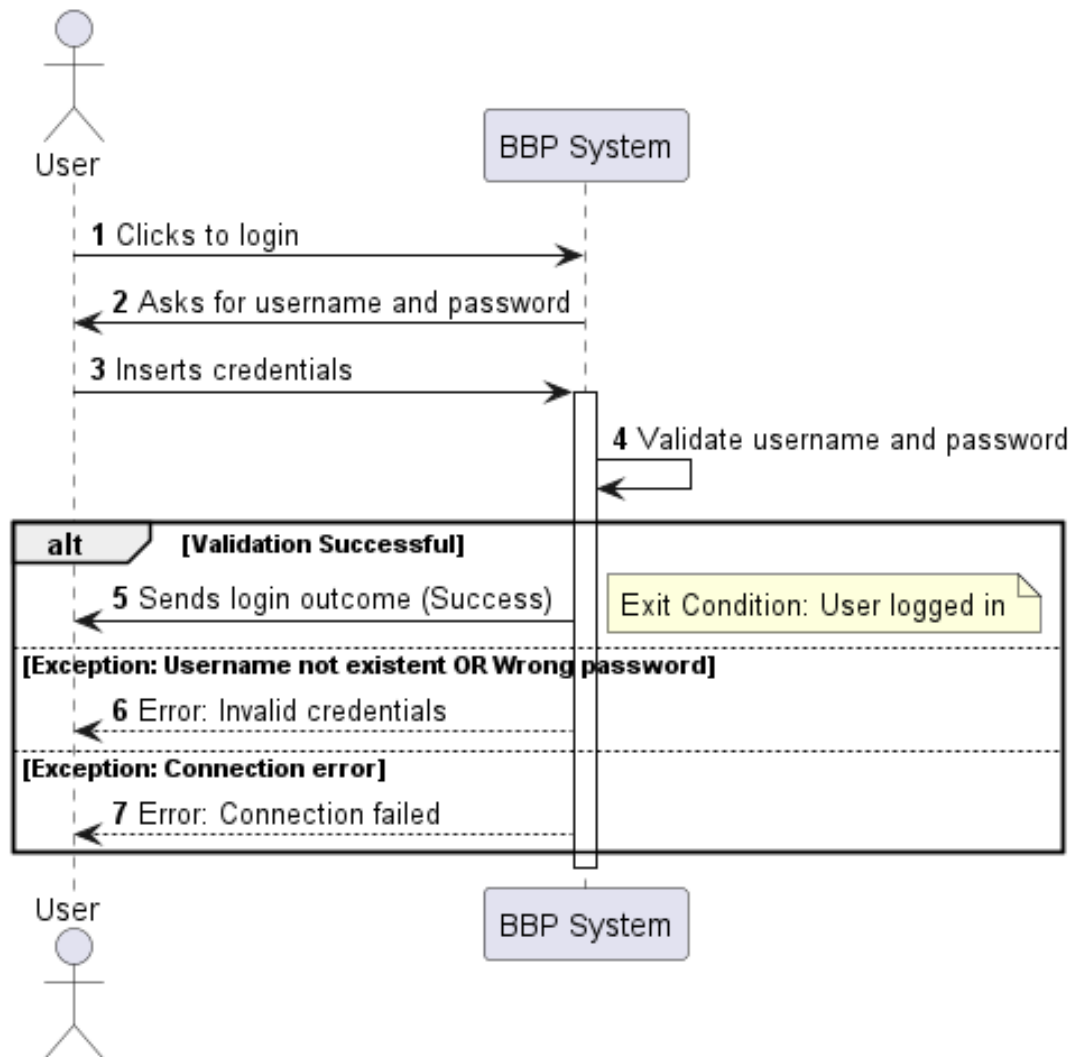


Figure 9: Use Case Diagram for UC2: User logs in the account

UC3. User manually adds bike trip

Actor	Registered user
Entry conditions	User is registered and logged in User is on "New trip" page
Event flow	1. User clicks on "Insert a bike trip" button 2. BBP asks to insert a name for the trip 3. User inserts name of the trip 4. BBP validates inserted name 5. User inserts traveled roads 6. BBP calculate and returns the complete traveled route 7. User confirms calculated route 8. BBP adds meteorological informations based on date and location of the trip 9. User inserts optional information, i.e obstacles 10. BBP calculates status based on inserted informations and shows to user 11. User clicks "Save trip" button 12. BBP saves trip, and informs user of outcome
Exit conditions	Inserted trip is saved
Exceptions	BBP can't calculate a route based on inserted roads User already saved a route with the same name

Table 8: Use Case UC3: User manually adds bike trip

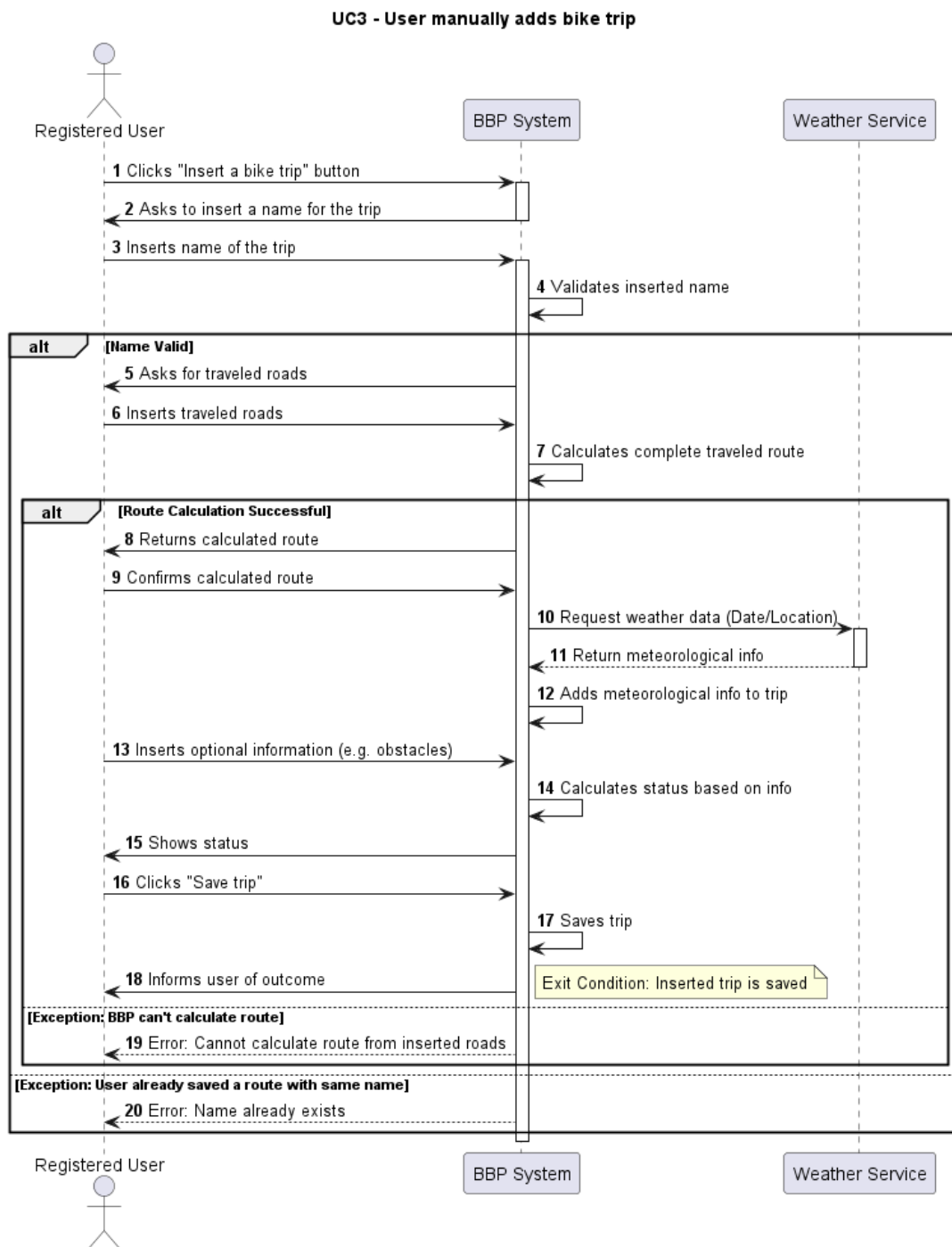


Figure 10: Use Case Diagram for UC3: User manually adds bike trip

UC4. User automatically records bike trip

Actor	Registered user
Entry conditions	User is registered and logged in User is on "New trip" page
Event flow	1. User clicks to start automatic recording mode 2. BBP starts taking data from user's phone GPS 3. BBP recognize user is cycling, based on his velocity from sensors 4. BBP traces position and encountered obstacles with user's phone sensors and GPS 5. BBP recognize user stopped cycling 6. User finished automatic recording 7. BBP asks user to confirm or correct recorded data 8. User optionally corrects data and click confirm 9. BBP adds meteorological informations based on date and location of the trip 10. BBP calculate status based on recorded and inserted data and shows to user 11. User insert a name for the trip and clicks "Save trip" button 12. BBP saves trip, and informs user of outcome
Exit conditions	Recorded trip is saved
Exceptions	GPS signal lost Sensor permissions negated User already saved a route with the same name User discards recorded trip

Table 10: Use Case UC4: User automatically records bike trip

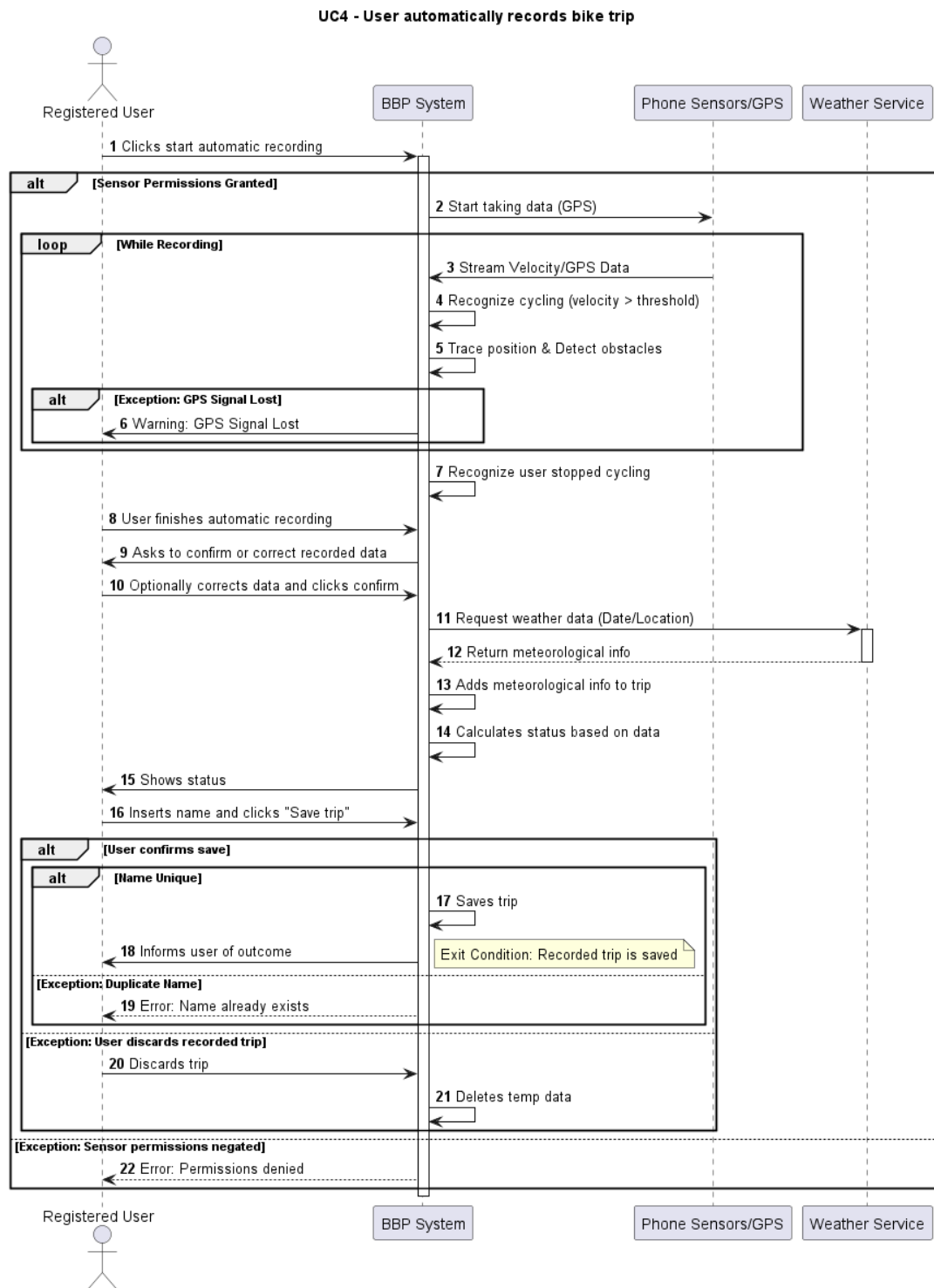


Figure 11: Use Case Diagram for UC4: User automatically records bike trip

UC5. User views his bike trip history

Actor	Registered user
Entry conditions	User is registered and logged in User is on "Your trips" page
Event flow	1. User opens on "Bike trip history" section 2. BBP loads history of trips of user 3. User visualizes his trips
Exit conditions	User closes "Bike trip history" section
Exceptions	User does not have any saved trips

Table 11: Use Case UC5: User views his bike trip history

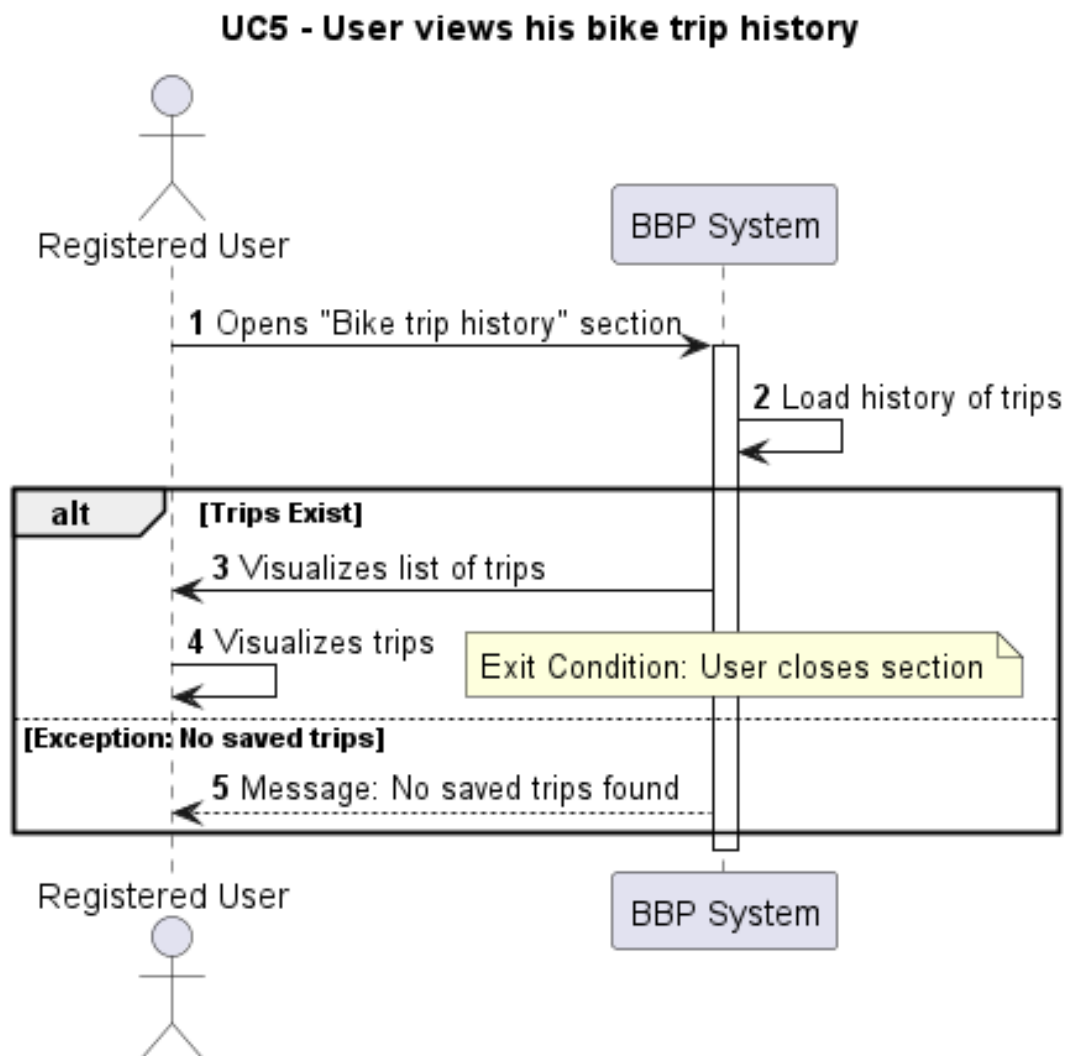


Figure 12: Use Case Diagram for UC5: User views his bike trip history

UC6. User views his bike trip statistics

Actor	Registered user
Entry conditions	User is registered and logged in User selected a specific trip from history
Event flow	1. User clicks on "View statistics" button 2. BBP loads complessive statistics of traveled bike trips, and statistics for each trip 3. User clicks on a specific trip 4. BBP visualizes in-depth statistics for that trip
Exit conditions	User closes "Trips Statistics" section
Exceptions	User does not have any trip to calculate statistics on

Table 12: Use Case UC6: User views his bike trip statistics

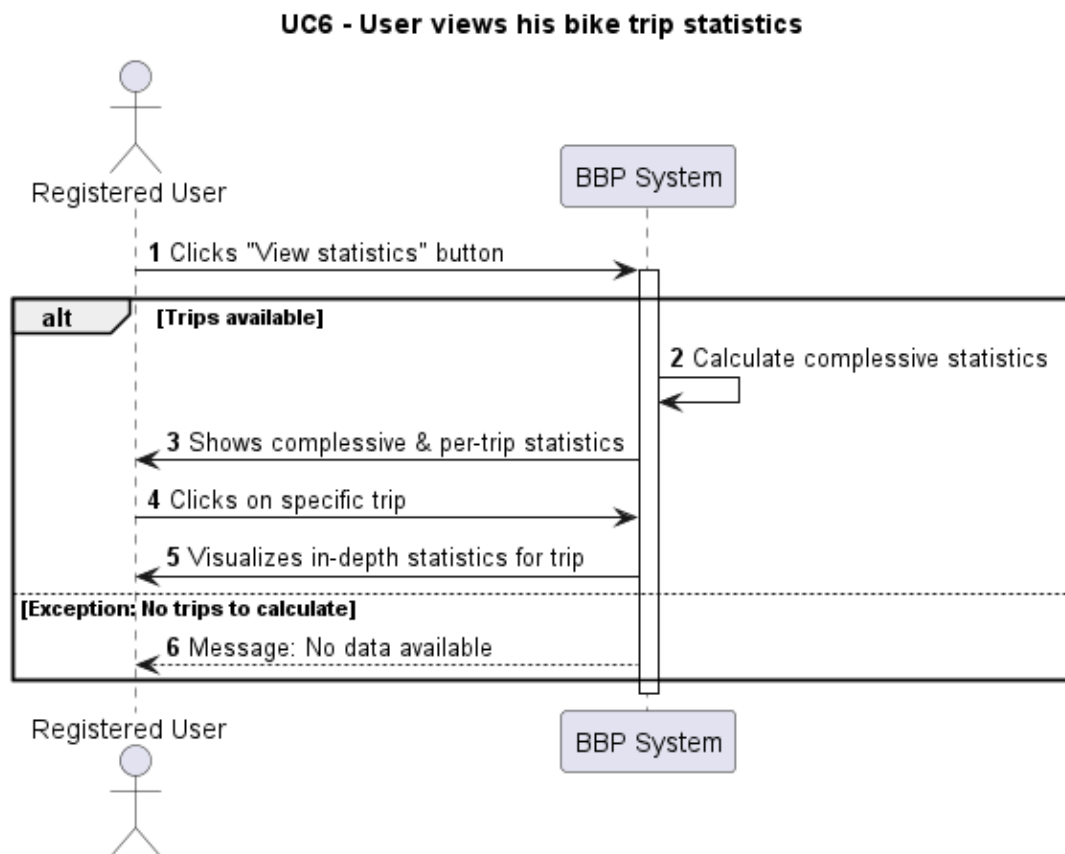


Figure 13: Use Case Diagram for UC6: User views his bike trip statistics

UC7. User publishes recorded trip

Actor	Registered user
Entry conditions	User is registered and logged in User selected a specific trip from history
Event flow	1. User click on "Publish trip" 2. BBP ask confirm to user if information are correct and if he is sure to make the trip public 3. User confirms the intention to publish 4. BBP publishes the trip, making it visible to all users
Exit conditions	User's trip is published and visible by all users
Exceptions	Connection error while publishing

Table 13: Use Case UC7: User publishes recorded trip

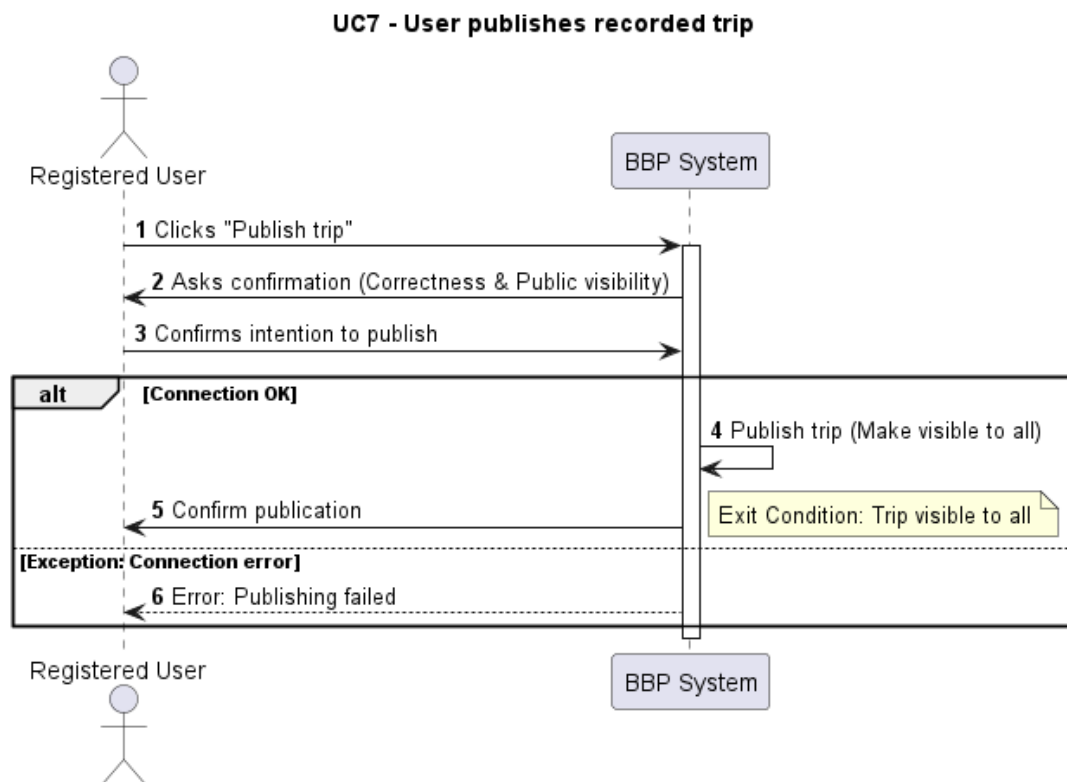


Figure 14: Use Case Diagram for UC7: User publishes recorded trip

UC8. User modify one of his saved trips

Actor	Registered user
Entry conditions	User is registered and logged in User selected a specific trip from history
Event flow	1. User click on "Modify trip" 2. BBP allows users to modify saved data about trip 3. User modifies trip data 4. BBP validate changes of trip data 5. User confirms the changes of trip data 6 BBP recomputes trip statistics based on modified data 7. BBP modifies trip data on database 8. If the trip is public BBP propagates changes for every user
Exit conditions	Trip data is modified
Exceptions	Inserted data is not valid Connection error while propagating changes of public trip

Table 14: Use Case UC8: User modifies one of his saved trips

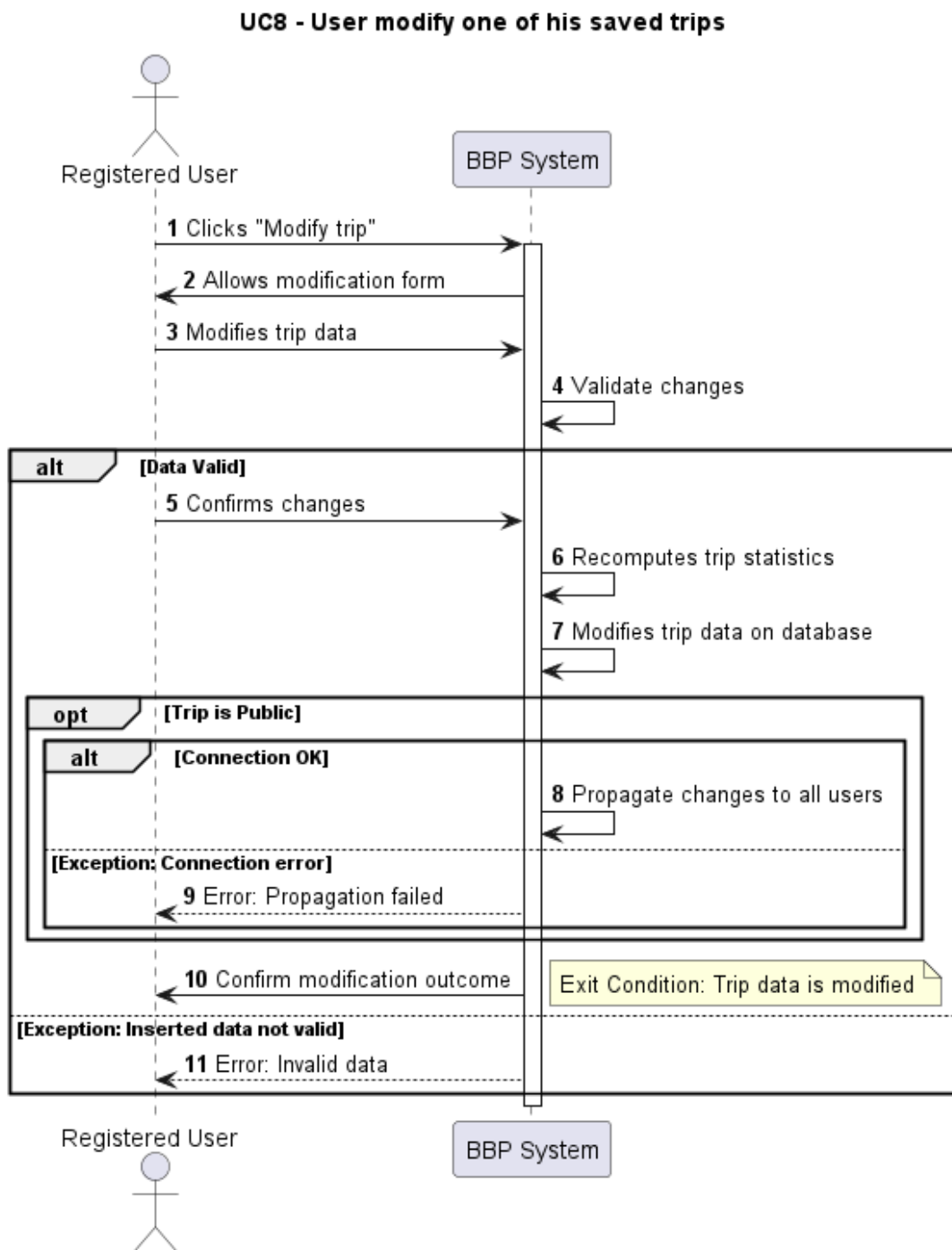


Figure 15: Use Case Diagram for UC8: User modifies one of his saved trips

UC9. User views possible bike routes on map

Actor	Registered user Unregistered user
Entry conditions	User is on "Path search" section
Event flow	1. BBP asks user to insert start and end point of route 2. User inserts start and end point 3. BBP visualizes every routes, with related score, between selected points
Exit conditions	Paths are visualized
Exceptions	Mapping service unavailable Connection error while retrieving paths No route found between the selected points

Table 15: Use Case UC9: User views possible bike routes on map

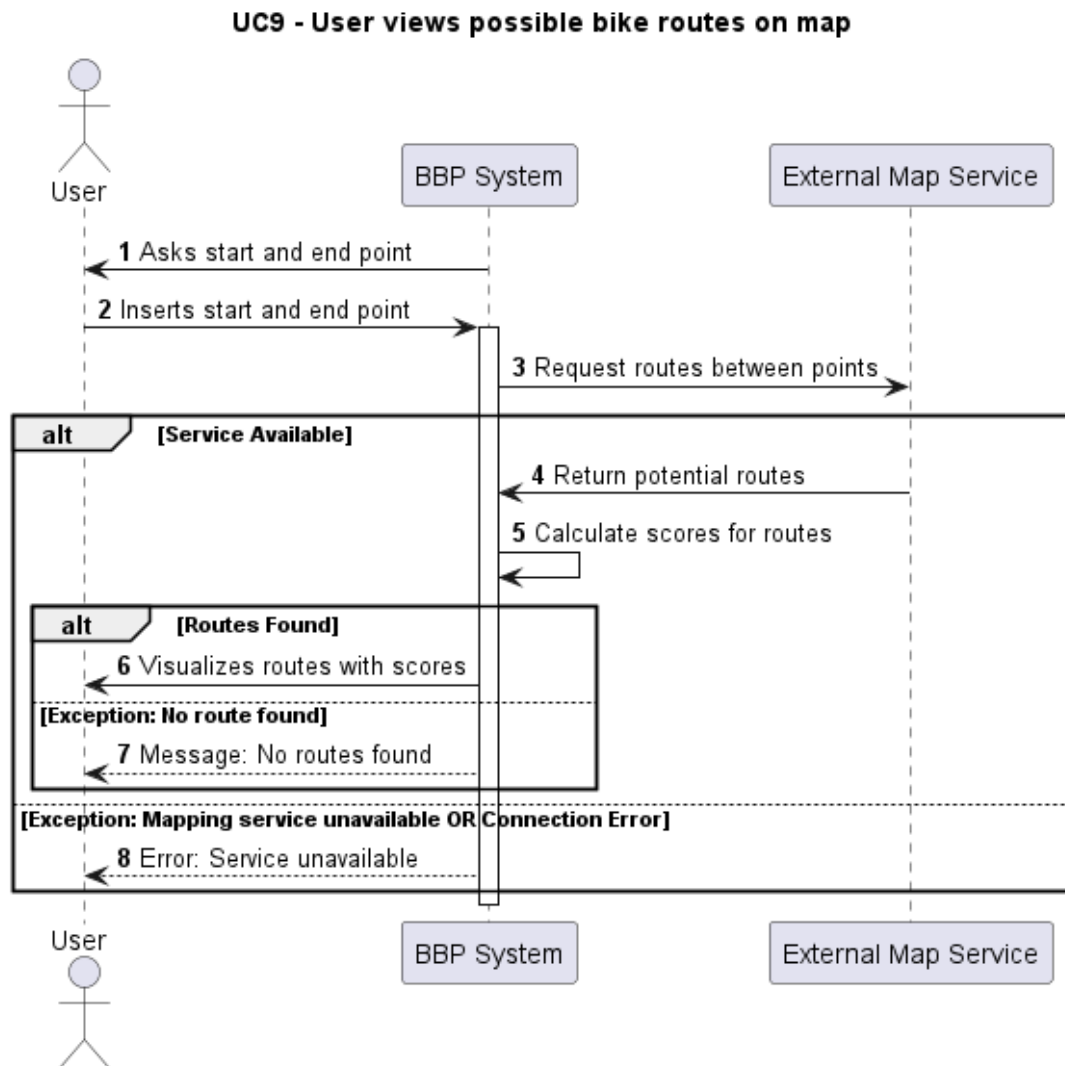


Figure 16: Use Case Diagram for UC9: User views possible bike routes on map

UC10. User views additional information about selected path on map

Actor	Registered user Unregistered user
Entry conditions	User is on "Path search" section visualizing paths between 2 points
Event flow	1. User clicks on a specific path on the map 2. BBP shows details about trip, i.e statistics, distance, obstacles etc..
Exit conditions	Additional informations of a trip are visualized
Exceptions	

Table 16: Use Case UC10: User views additional information about selected path on map

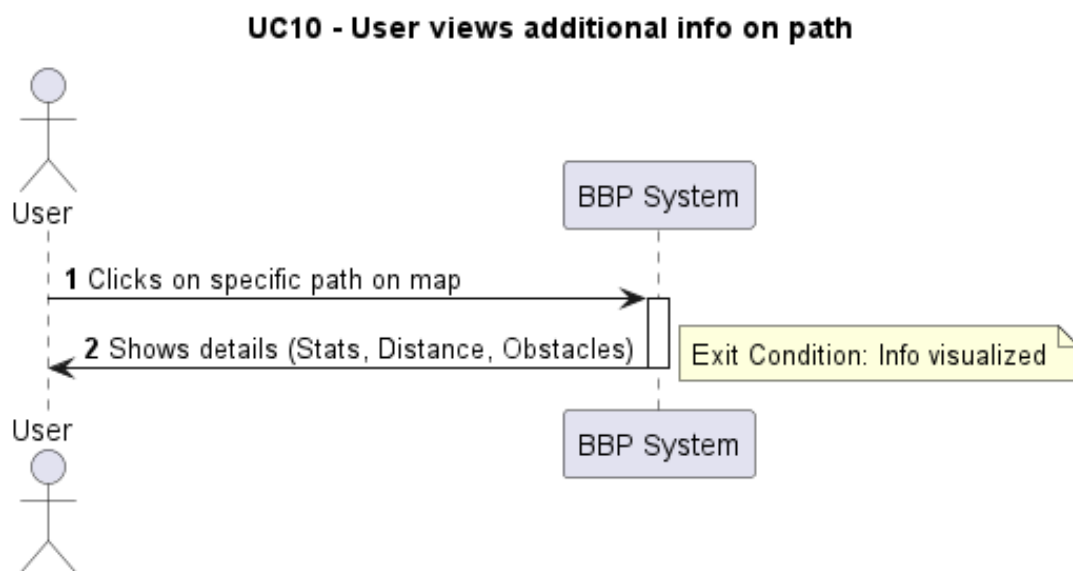


Figure 17: Use Case Diagram for UC10: User views additional information about selected path on map

UC11. User start navigation on selected bike path

Actor	Registered user Unregistered user
Entry conditions	User selected a path on the map
Event flow	1. User click to open route on a third-party mapping service 2. BBP opens third-party mapping service, giving it coordinates of start and end point
Exit conditions	Additional informations of a trip are visualized
Exceptions	third-party mapping service not available

Table 17: Use Case UC11: User start navigation on selected bike path

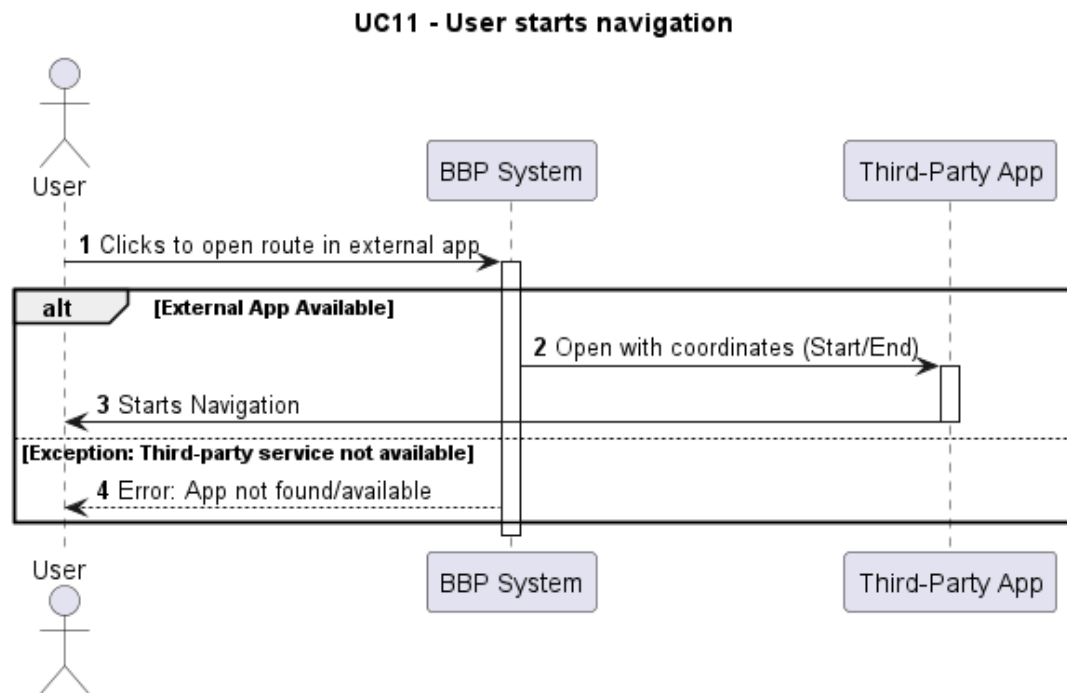


Figure 18: Use Case Diagram for UC11: User start navigation on selected bike path

UC12. User modifies profile data

Actor	Registered user
Entry conditions	User is registered and logged in User is on profile page of BPP
Event flow	1. User clicks on modify profile data 2. User insert new personal data 3. BBP validates new personal data 4. User confirms changes on personal data 5. BBP update user personal data
Exit conditions	User personal data are modified
Exceptions	New username already taken New password not valid Connection error while modifying profile data

Table 18: Use Case UC12: User modifies profile data

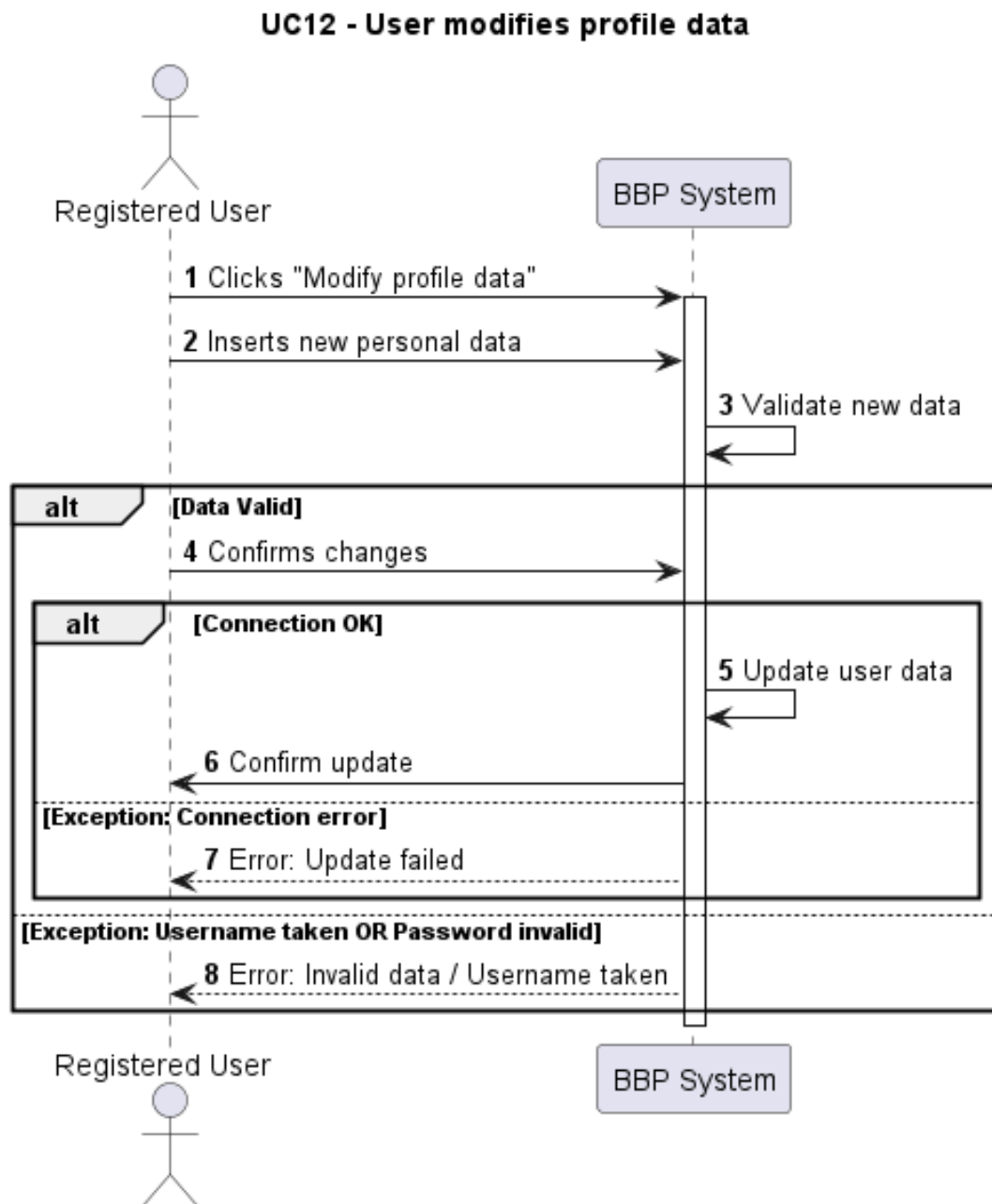


Figure 19: Use Case Diagram for UC12: User modifies profile data

UC13. User logs out the account

Actor	Registered user
Entry conditions	User is registered and logged in User is on profile page of BBP
Event flow	1. User clicks on log out button 2. BBP terminate access of user on the account 3. BBP returns user to log in page
Exit conditions	User is disconnected from the account
Exceptions	

Table 19: Use Case UC13: User logs out the account

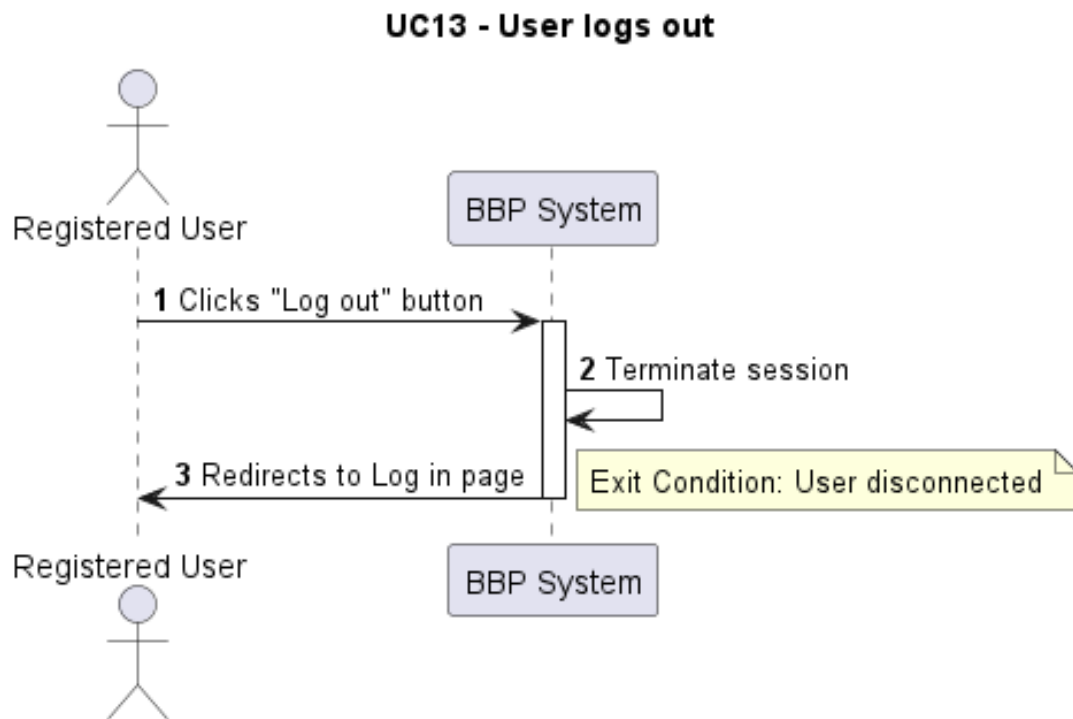


Figure 20: Use Case Diagram for UC13: User logs out the account

UC14. User deletes the account

Actor	Registered user
Entry conditions	User is registered and logged in User is on profile page of BPP
Event flow	1. User clicks on delete account button 2. BBP asks for confirmation 3. User confirms account deletion 4. BBP deletes user account and all associated data 5. BBP returns user to home page
Exit conditions	User is disconnected from the account and account is deleted
Exceptions	Connection error while deleting account

Table 20: Use Case UC14: User deletes the account

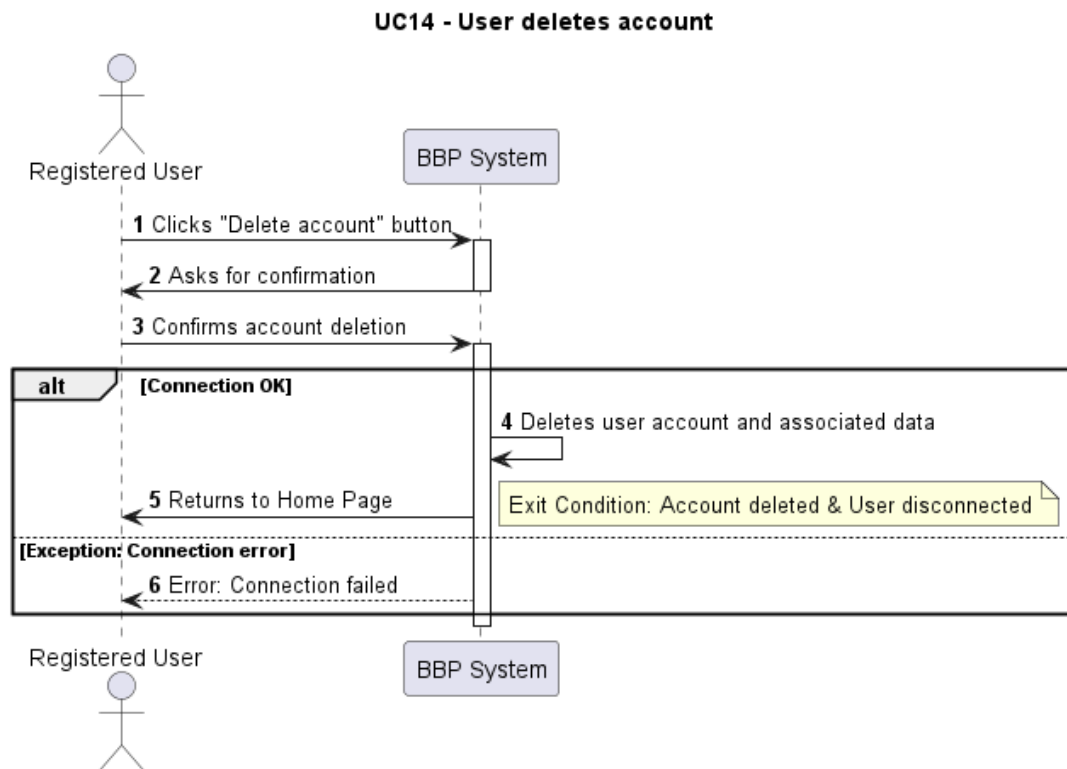


Figure 21: Use Case Diagram for UC14: User deletes the account

UC15. User recovers password of the account

Actor	Registered user
Entry conditions	User is on login page of BBP
Event flow	1. User clicks on "Forgotten password" button 2. BBP asks for email of the account 3. User inserts email associated to the account 4. BBP sends an OTP to the inserted email 5. User inserts received OTP on BPP 6. BBP validates inserted OTP 7. User inserts new password 8. BBP validates new password 9. User confirms new password 10. BBP updates password of the account 11. BBP returns user to login page
Exit conditions	Password is successfully reset
Exceptions	Connection error while resetting password Invalid OTP New password not valid

Table 21: Use Case UC15: User recovers password of the account

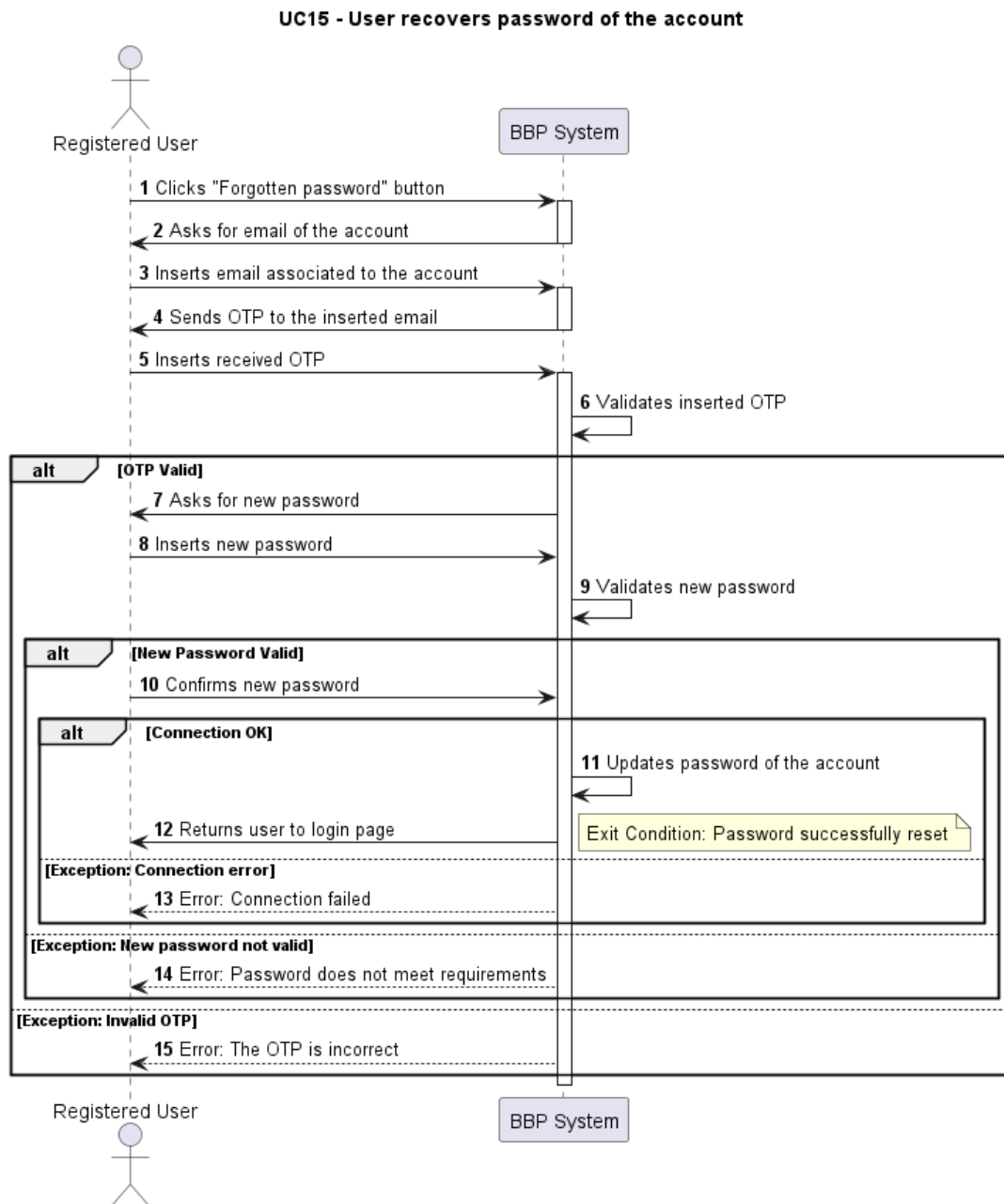


Figure 22: Use Case Diagram for UC15: User recovers password of the account

UC16. User deletes a recorded trip

Actor	Registered user
Entry conditions	User is registered and logged in User selected a specific trip from history
Event flow	1. User clicks on "Delete trip" button 2. BBP asks for confirmation 3. User confirms deletion 4. BBP deletes trip from database 5. If the trip is public BBP propagates deletion for every user
Exit conditions	Trip is successfully deleted
Exceptions	Connection error while deleting trip

Table 22: Use Case UC16: User deletes a recorded trip

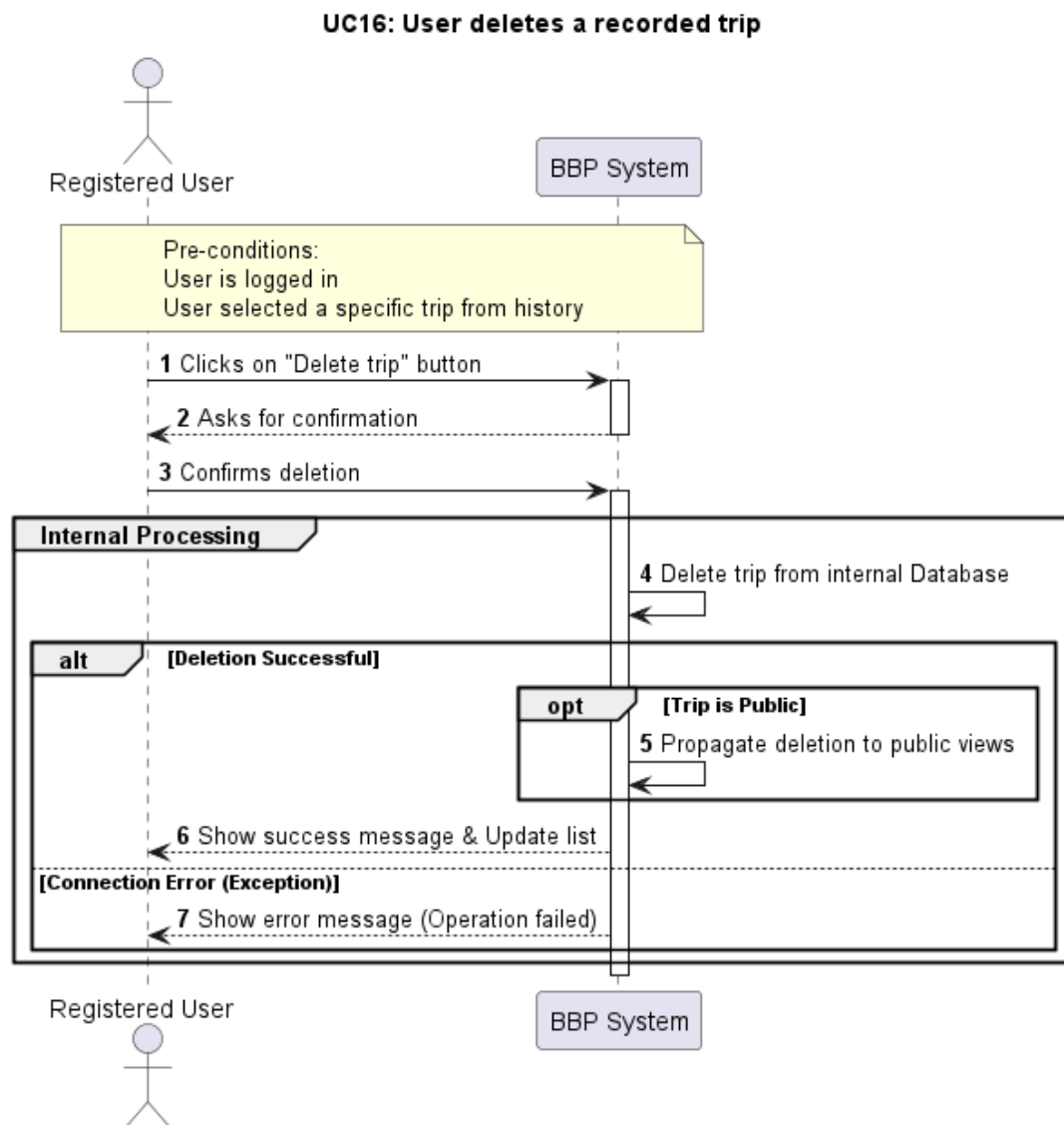


Figure 23: Use Case Diagram for UC16: User deletes a recorded trip

3.2.5 Mapping on requirements

Table 23: Use Case Mapping on Requirements

Use case	Requirements
UC1	R1. System accepts account registration data R2. System rejects account registrations with duplicate emails or usernames R3. System sends a verification email with an OTP to the email address provided during registration R4. System must verify OTP code before confirming registration
UC2	R5. System allows users to login to their account by authenticating via email (or username) and password R8. System allows users to modify/recover their password via email verification
UC3	R19. System allows users to manually record trips by inserting starting and ending locations, names of roads in traversal order, start and end times R20. System allows users to specify road status (optimal, medium, sufficient, requires maintenance) during manual recording R21. System allows users to specify obstacle locations during manual recording R22. System persistently stores recorded trips R23. System ensures only author of recorded trips can access unpublished trip records R24. System automatically calculates statistics for each recorded trip (distance, average speed, duration, elevation, average slope) R25. System calculates statistics immediately after trip is saved R26. System calculates aggregate statistics across all of user's recorded trips R27. System recomputes aggregate statistics upon modification of user's trip record (add/delete/edit trip) R28. System persistently stores computed statistics with corresponding trip data R30. System queries an external meteorological service for weather data to enrich trip data after completion R31. System stores a notice in trip data when weather data is unavailable

Table 23: Use Case Mapping on Requirements

Use case	Requirements
UC4	R10. System allows users to start automated trip recording R11. System requests necessary sensor and GPS permissions when automated recording is started R12. System begins automated path and sensor tracking, after recording initiation, only when user has started biking R13. System algorithmically detects if a user has started biking from speed data R14. System collects GPS coordinates during automated trip recording R15. System reconstructs traveled path from collected GPS coordinates R16. System collects data from mobile device's accelerometer and gyroscope during automated trip recording R17. System algorithmically interprets collected sensor data to detect potential obstacles along the path R18. System allows users to manually stop automated trip recording R22. System persistently stores recorded trips R23. System ensures only author of recorded trips can access unpublished trip records R24. System automatically calculates statistics for each recorded trip (distance, average speed, duration, elevation, average slope) R25. System calculates statistics immediately after trip is saved R26. System calculates aggregate statistics across all of user's recorded trips R27. System recomputes aggregate statistics upon modification of user's trip record (add/delete/edit trip) R28. System persistently stores computed statistics with corresponding trip data R29. System infers road status from detected obstacle density and movement pattern analysis when trip data is recorded automatically R30. System queries an external meteorological service for weather data to enrich trip data after completion R31. System stores a notice in trip data when weather data is unavailable
UC5	R23. System ensures only author of recorded trips can access unpublished trip records R26. System calculates aggregate statistics across all of user's recorded trips R32. System displays a list of user's recorded trips showing name, date, start and end locations R33. System allows users to view detailed trip data for any recorded trip, including route map, statistics and weather data
UC6	R24. System automatically calculates statistics for each recorded trip (distance, average speed, duration, elevation, average slope) R26. System calculates aggregate statistics across all of user's recorded trips R28. System persistently stores computed statistics with corresponding trip data R30. System queries an external meteorological service for weather data to enrich trip data after completion R31. System stores a notice in trip data when weather data is unavailable R33. System allows users to view detailed trip data for any recorded trip, including route map, statistics and weather data

Table 23: Use Case Mapping on Requirements

Use case	Requirements
UC7	R38. System allows registered users to publish their recorded trips, making them publicly available R39. System requires the user to correct and confirm data before publishing, if it was acquired automatically R40. System allows users to add/remove obstacle markers during data correction R41. System allows users to modify road status during data correction
UC8	R27. System recomputes aggregate statistics upon modification of user's trip record (add/delete/edit trip) R34. System allows users to manually edit trip data for any of their recorded trips R36. System propagates modifications and deletions of trips to their published versions, if present
UC9	R42. System allows all users to view published bike paths between specified start and end points R43. System identifies all published bike paths that connect specified start and end points R44. System notifies user when no published paths exist between the specified points R45. System calculates a score for each identified bike path based on road status, route effectiveness (distance and directness in reaching destination), amount of obstacles, path difficulty (slope) R46. System displays all identified paths on a map, with the highest scoring path highlighted and the others with reduced opacity R47. System displays score values for each path R48. System initially displays a limited amount of the identified paths, providing the option to view all paths R49. System allows users to select a displayed path R50. System allows users to view additional information about selected path (status, obstacles) R51. System allows users to export selected path to compatible third-party navigation applications
UC10	R49. System allows users to select a displayed path R50. System allows users to view additional information about selected path (status, obstacles) R51. System allows users to export selected path to compatible third-party navigation applications
UC11	R49. System allows users to select a displayed path R51. System allows users to export selected path to compatible third-party navigation applications
UC12	R7. System allows users to modify their account information (username and personal data) R8. System allows users to modify/recover their password via email verification
UC13	R6. System allows users to logout of their account
UC14	R9. System allows users to delete their account, with all the associated data

Table 23: Use Case Mapping on Requirements

Use case	Requirements
UC15	R8. System allows users to modify/recover their password via email verification
UC16	R27. System recomputes aggregate statistics upon modification of user's trip record (add/delete/edit trip) R35. System allows users to delete any of their recorded trips R36. System propagates modifications and deletions of trips to their published versions, if present

3.3 Performance Requirements

Number of users

The main performance requirement of the system is the ability to support a large number of users. Given that the number of recreational and professional cyclists is large: a survey in 2022 shows that around 37% of people in Italy state that they go cycling at least once a week, and in the european countries where the survey was taken the percentages of cyclists go from 25% (Canada) to 65% (Netherlands). Therefore, as the community of cyclists grows, we can expect a rapidly growing number of users; in light of this the scalability of the system is critical, as it must be able to accommodate increasing numbers of users with minimal performance loss.

Data Storage

Another critical aspect is the data storing capability of the system. For each bike trip record, the system must store path information and generated statistics: as the number of users and the number of records saved by each user grows, so will the amount of storage needed; therefore, the system's storage capabilities must be easily scalable proportionally to the number of users and the amount of records each user has saved.

Response Time

There are no strict time response requirements, therefore the system should handle all requests within 3 seconds.

3.4 Design Constraints

3.4.1 Standards Compliance

GDPR: the system manages personal user data, therefore it must comply with privacy and data treatment regulations; to work in the EU, it must comply with the EU GDPR.

3.4.2 Hardware Limitations

The BBP system is accessed exclusively through a mobile application. The user must have a mobile device with a compatible operating system (Android or iOS), and in order to utilize the automated recording function of BBP it must have a GPS module, an accelerometer and a gyroscope. The device must have the capability to maintain a stable internet connection.

3.4.3 Any Other Constraint

The mobile device should have a compatible third-party navigation application (e.g. Google Maps) if the user wants to utilize the navigation functionality of the path viewer.

3.5 Software System Attributes

3.5.1 Reliability

The BBP system main functionalities are about data management, which aren't to be considered critical, as in case of failure the operation can be repeated. The automatic tracking of trips is more sensitive, as in case of interruption some of the tracking data may be lost or incomplete. Therefore, a failure rate between 1% and 0.1% (between common systems and more high-quality reliable systems) would be ideal.

3.5.2 Availability

The BBP system should strive to give the best possible experience to its users, however it does not offer critical functionalities that require it to be constantly available. Therefore, an availability of 99% (with around 4 days of downtime per year) would be acceptable to guarantee that the users are satisfied with the service.

3.5.3 Security

The BBP system stores personal user information, and has communication between the system and the various users' mobile devices. Therefore, it should assure that data is private and encrypted when information is exchanged through the internet: a secure encrypted protocol such as HTTPS should be used. Furthermore, all information stored by BBP should be encrypted, and stored passwords should be hashed and salted in order to ensure privacy.

3.5.4 Maintainability

The BBP system should be appropriately divided into different modules based on the provided functionalities. The system should be built with ease of maintenance and scalability in mind, especially considering that the number of users could grow over time; the design should allow maintenance of single modules without impacting the rest of the system. Moreover, every implemented functionality should be rigorously and extensively documented.

3.5.5 Portability

The BBP software is a mobile application which does not have strict software or hardware restrictions. Therefore, to ensure a higher portability, the system should be developed for both Android and iOS if possible, in order to be available on the majority of modern smartphones. Considering budget constraints the application could either be developed separately for each platform, which is more costly in terms of development and maintenance, or using cross-platform development tools, which would reduce the development cost.

4 Formal Analysis Using Alloy

This section regards the formal analysis of the BBP system using the Alloy specification language. The analysis focalizes on the management of the trips recorded by the various users, showing the evolution of the system after a trip is recorded, modified, verified, published, or deleted. This analysis is meant to show how the various components of the system interact while it is carrying out one of its core functionalities, and verifies that all the steps of the trip lifecycle have been correctly defined. All predicates shown in the Alloy code below have been found to be consistent.

4.1 Alloy Code

```

open util/integer
open util/ordering[TimeStamp]

// SIGNATURES -----

sig User {}

sig RegisteredUser extends User {
  var records: disj set Trip,
  generates: disj AggregateStatistics,
  has: disj MobileDevice
}

sig Trip {
  mode: RecordingMode,
  var isPublished: Bool,
  var isVerified: Bool,
  generates: disj Statistics,
  traverses: disj Path,
  enrichedBy: disj lone MeteorologicalData,
  startTime: TimeStamp,
  endTime: TimeStamp
} {
  lt[startTime, endTime]
}

sig AggregateStatistics {
} {
  this in RegisteredUser.generates
}

sig Statistics {
} {
  this in Trip.generates
}

sig Path {
  composedBy: seq PathSegment
} {
  this in Trip.traverses and
  #composedBy > 0
}

```

```
}

sig PathSegment {
  contains: disj set Obstacle,
  starts: Location,
  ends: Location,
  status: RoadStatus
} {
  this in Path.composedBy.elems and
  starts != ends
}

sig Obstacle {
  positioned: Location
} {
  this in PathSegment.contains
}

sig MeteorologicalData {
} {
  this in Trip.enrichedBy
}

sig MobileDevice {
  contains: disj some Sensor
} {
  this in User.has
}

sig Sensor {
  type: SensorType
} {
  this in MobileDevice.contains
}

sig Location {
} {
  this in PathSegment.starts or
  this in PathSegment.ends or
  this in Obstacle.positioned
}

sig TimeStamp {
} {
  this in Trip.startTime or
  this in Trip.endTime
}

enum RecordingMode {MANUAL, AUTOMATED}

enum RoadStatus {OPTIMAL, MEDIUM, SUFFICIENT, REQUIRES_MAINTENANCE}
```

```
enum SensorType {GPS, GYRO, ACCELEROMETER}

enum Bool {TRUE, FALSE}

// FACTS -----

// A trip must be verified in order to be published
fact PublishedTripMustBeVerified {
    always(
        all t: Trip |
            (t.mode = AUTOMATED and t.isPublished = TRUE) implies
                t.isVerified = TRUE
    )
}

// When trips are recorded, they are private
fact TripsStartNotPublished {
    all t: Trip |
        t.isPublished = FALSE and
        after always (
            t.isPublished = TRUE
            implies
                once t.isPublished = FALSE
        )
}

// If an trip is recorded in automated mode, it is not verified
// until the user corrects the data
fact TripsStartNotVerifiedIfAutomated {
    all t: Trip |
        t.mode = AUTOMATED implies (
            t.isVerified = FALSE and
            after always (
                t.isVerified = TRUE
                implies
                    once t.isVerified = FALSE
            )
        )
}

// Manually inserted trips do not need to be verified
fact ManualTripsVerified {
    always (
        all t: Trip |
            t.mode = MANUAL implies t.isVerified = TRUE
    )
}

// Path segments that compose a path must be connected
```

```

fact PathSegmentsAreConnected {
  all p: Path |
    all i: Int |
      (i in p.composedBy.inds and i != p.composedBy.lastIdx)
      implies
        p.composedBy[i].ends = p.composedBy[add[i,1]].starts
}

// A user cannot have two trips that take place at the same time
fact NoConcurrentTrips {
  all u: RegisteredUser |
    all disj t1, t2: Trip |
      (t1 in u.records and t2 in u.records)
      implies
        (lte[t1.endTime, t2.startTime] or lte[t2.endTime, t1.startTime])
}

// All user's mobile devices have GPS functionalities
fact AllUsersHaveGPS {
  all u: RegisteredUser |
    one s: Sensor | s.type = GPS and s in u.has.contains
}

// Trips can be owned by only one user
fact TripsOwnedByOnlyOneUser {
  all t: Trip, u: RegisteredUser |
    t in u.records implies always(
      no u1: RegisteredUser |
        u1 != u and t in u1.records
    )
}

// PREDICATES -----

pred recordTrip[u: RegisteredUser, t: Trip, m: RecordingMode] {
  no u2: RegisteredUser | t in u2.records
  t.mode = m

  u.records' = u.records + t

  t.isPublished' = FALSE

  all u1: RegisteredUser - u | u1.records' = u1.records

  all t1: Trip - t | {
    t1.isPublished' = t1.isPublished
    t1.isVerified' = t1.isVerified
  }
}

```

```

run recordTrip

pred verifyTrip[u: RegisteredUser, t: Trip] {
  t in u.records
  t.isVerified = FALSE

  t.isVerified' = TRUE

  -- Frame Conditions
  t.isPublished' = t.isPublished

  all u1: RegisteredUser | u1.records' = u1.records

  all t1: Trip - t | {
    t1.isPublished' = t1.isPublished
    t1.isVerified' = t1.isVerified
  }
}

run verifyTrip

pred publishTrip[u: RegisteredUser, t: Trip] {
  t in u.records
  t.isPublished = FALSE

  (t.mode = AUTOMATED) implies (t.isVerified = TRUE)

  t.isPublished' = TRUE

  -- Frame Conditions
  t.isVerified' = t.isVerified

  all u1: RegisteredUser | u1.records' = u1.records

  all t1: Trip - t | {
    t1.isPublished' = t1.isPublished
    t1.isVerified' = t1.isVerified
  }
}

run publishTrip

pred deleteTrip[u: RegisteredUser, t: Trip] {
  t in u.records

  u.records' = u.records - t

  -- Frame Conditions
  all u1: RegisteredUser - u | u1.records' = u1.records
}

```

```

all t1: Trip - t | {
  t1.isPublished' = t1.isPublished
  t1.isVerified' = t1.isVerified
}

run deleteTrip

pred showScenario {
  some disj u1, u2: RegisteredUser, disj t1, t2: Trip | {
    recordTrip[u1, t1, AUTOMATED]
    after verifyTrip[u1, t1]
    after after publishTrip[u1, t1]
    after after after recordTrip[u2, t2, MANUAL]
    after after after after deleteTrip[u2, t2]
  }
}

run showScenario for 5 but 3 Int, 6 TimeStamp

```

4.2 Simulations

The simulations shown refer to the predicate 'showScenario'. They show the evolution of the world after each defined operation is performed (recording, verifying, publishing and deleting a trip); two different registered users are shown performing the operations so that it is apparent that they according to what would be expected.

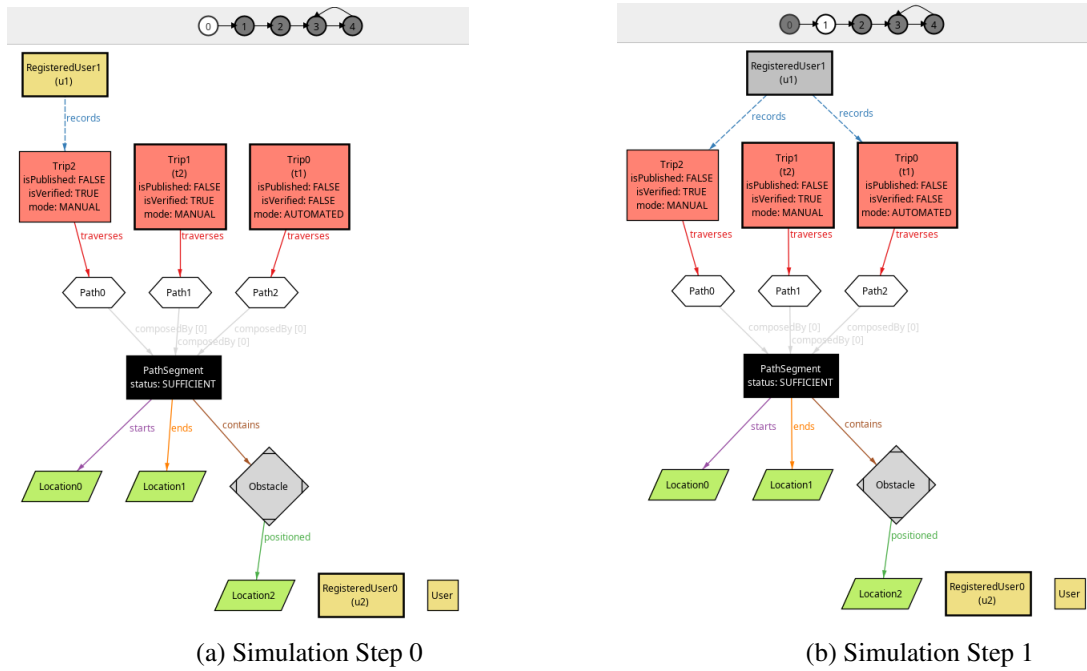


Figure 24: Simulation Steps 0-1

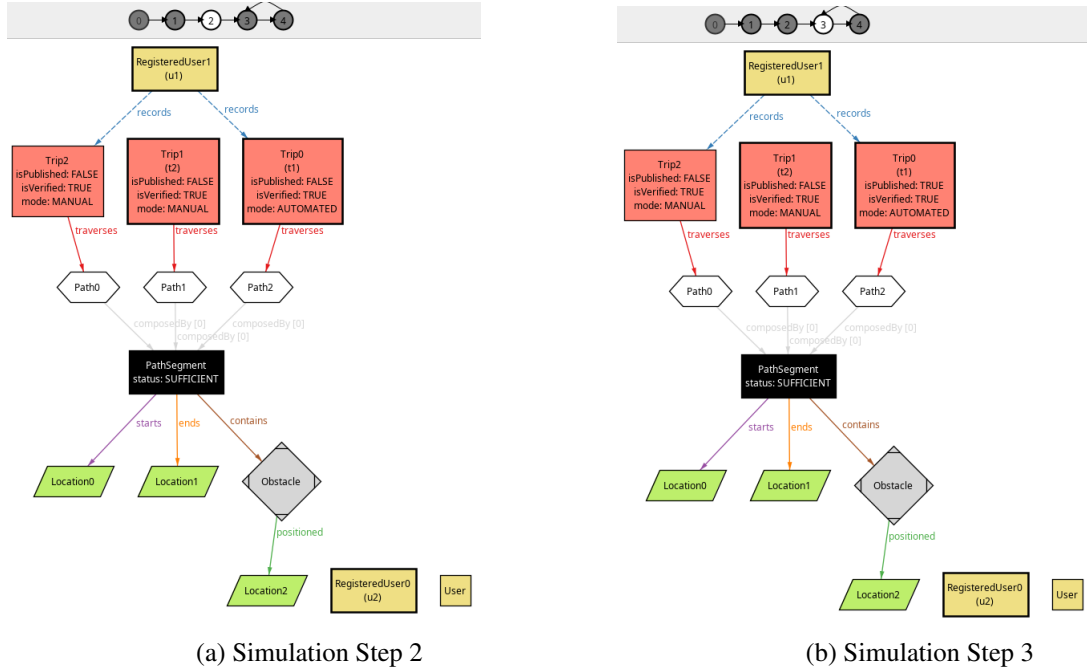


Figure 25: Simulation Steps 2-3

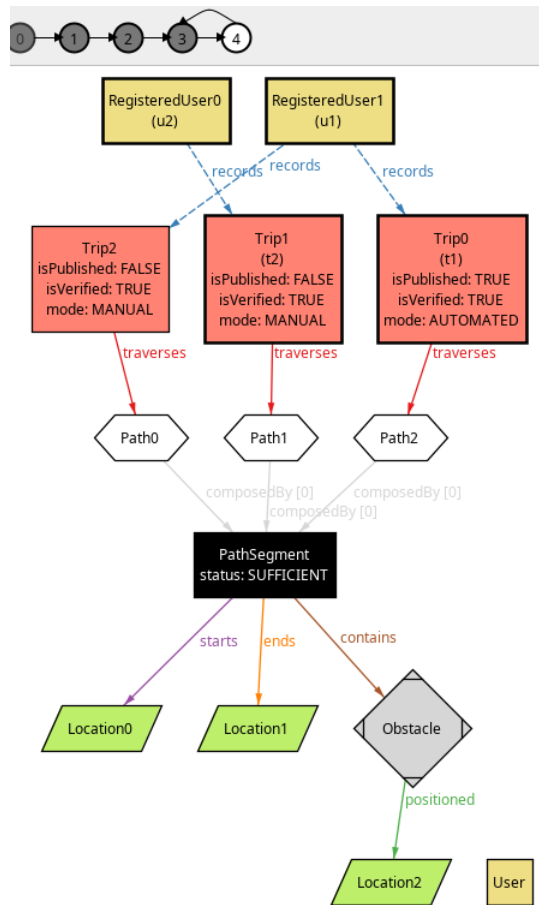


Figure 26: Simulation Step 4

5 Effort Spent

Member of group	Effort spent	
Bedini Simone	Introduction	9h
	Overall description	10h
	Specific requirements	11h
	Formal analysis	15h
	Reasoning	13h
Cantoli Angelo	Introduction	9h
	Overall description	11h
	Specific requirements	24h
	Formal analysis	2h
	Reasoning	11h

6 References

6.1 Reference Documents

- Assignment RDD AY 2025-2026.pdf
- Lecture slides and notes from the course Software Engineering 2 2025/2026
- Global Advisor-Cycling Across the World-2022 Report.pdf - <https://www.ipsos.com/sites/default/files/ct/news/documents/2022-05/Global>

6.2 Used Tools

The following tools have been used to create this document:

- **LaTeX**: for document preparation and typesetting.
- **PlantUML**: for creating UML diagrams included in the document.
- **GitHub**: for version control and collaboration on the document source files.
- **Alloy Analyzer**: for formal analysis and verification of system specifications.
- **VSCode**: as the main text editor for writing LaTeX code.